KENYA CERTIFICATE OF SECONDARY EDUCATION
COMPUTER PROJECT 2025 (451/3)

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VICTORY SCHOOL CLUB
System
VICTORY SCHOOL CLUB
System

SCHOOL CODE: /SCHOOL/

INDEX NUMBER: /INDEX NO/

PROJECT TITLE: Victory School Club Membership System

YEAR OF EXAMINATION: 2024

SCHOOL: /SCHOOL/

DECLARATION

7.1 STUDENT'S DECLARATION

I, [Student's Full Name], declare that this Computer Studies Project on the **Victory School Club Membership System** is my original work. All the information and documentation presented in this project are authentic and have been developed solely for the purpose of fulfilling the requirements of the Computer Studies examination. Any external sources referenced in this project have been appropriately cited. I understand the consequences of academic dishonesty, and I affirm that this project represents my independent efforts and knowledge acquired during the course. I take full responsibility for the content presented.

Signature: _____

Date: _____

7.2 SUPERVISOR DECLARATION

I, [Supervisor/Teacher's Full Name], hereby declare that I have supervised and guided [Student's Full Name] in the development of the **Victory School Club Membership System** for the Computer Studies Project. I have reviewed the project documentation and implementation, and I affirm that the work presented is consistent with the guidelines and expectations of the KNEC. I believe that the student has demonstrated a sound understanding of the concepts and skills covered in the Computer Studies curriculum. I take responsibility for providing necessary guidance and support throughout the project's development.

Signature: _____

Date: _____

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DEDICATION

I dedicate this Computer Studies Project on the **Victory School Club Membership System** to my parents. Your unwavering support, encouragement, and guidance have been the foundation of my academic journey. This project is a testament to the inspiration and motivation I have gained from your belief in my potential. I am deeply grateful for the sacrifices you've made and for always pushing me to strive for excellence. This work is a reflection of your influence and the positive impact you've had on my education.

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ACKNOWLEDGMENT

I would like to express my sincere appreciation to everyone who contributed to the successful completion of this Computer Studies Project on the **Victory School Club Membership System**. My heartfelt gratitude goes to my supervisor, **[Teacher's Name]**, for their invaluable guidance, constructive feedback, and unwavering support throughout the project. I am also grateful to my mentors for providing the necessary knowledge and inspiration that fueled my understanding of the subject matter. Special thanks to **[School's Name]** for their encouragement and assistance during challenging moments. Lastly, I extend my appreciation to my family and friends for their understanding, patience, and encouragement throughout this academic endeavor.

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CHAPTER 1:

INTRODUCTION

1.1 BACKGROUND INTRODUCTION

Victory School is a vibrant institution that offers a wide range of co-curricular activities, fostering the holistic development of its students beyond the formal curriculum. Among the many avenues for student engagement, clubs and societies play a central role in promoting teamwork, leadership, creativity, and academic excellence. These student-led organizations provide a platform for students to pursue shared interests and hobbies in various fields, such as debating, music, science, drama, and agriculture, among others.

To facilitate smooth operations and efficient management of these clubs, there is a pressing need for an automated system that can manage club memberships, track club activities, handle finances, and generate reports for better decision-making. As the number of clubs and their members continue to increase, it becomes imperative for Victory School to streamline the management of student memberships and club finances to ensure that each club is effectively supported and well-organized.

This Computer Studies Project aims to address the administrative challenges faced by Victory School by developing a comprehensive **Victory School Club Membership System**. This system is designed to automate and manage crucial aspects of club operations, including member registration, categorization of roles, fee management, event tracking, financial allocations, and more. With this system in place, Victory School will be able to better support its clubs, enhance student participation, and ensure that the available resources are efficiently utilized for the benefit of the students and the school community as a whole.

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CHAPTER 2:

SYSTEM ANALYSIS

2.1 PROBLEM DEFINITION

Victory School faces significant challenges in the management of its co-curricular clubs, which are an integral part of the students' academic and personal development. The current manual processes for handling club memberships, activities, finances, and reports are cumbersome, error-prone, and inefficient. Managing student registrations, categorizing members into their respective roles (Regular or Executive), and tracking financial transactions like registration fees and revenue from club activities is currently done manually, which is time-consuming and prone to inaccuracies.

The lack of a centralized system for storing and managing club-related data leads to inefficiencies in the retrieval and analysis of information. For instance, club members' roles and participation may not be accurately tracked, and the process of handling registration payments is disorganized. Additionally, tracking financial contributions and allocating funds for activities, outings, and club savings are done manually, often leading to discrepancies and delays.

Furthermore, generating financial summaries, reports on club activities, and auditing club expenses is an arduous process without an automated system. As a result, the school's administrative staff faces increased workload, and decision-making related to the funding and management of clubs lacks the data-driven precision needed to ensure efficient use of resources.

In light of these challenges, the need for an automated **Victory School Club Membership System** becomes evident. This system is intended to streamline and automate the management of club memberships, financial transactions, event tracking, and reporting, ensuring better accuracy, efficiency, and transparency in handling the school's co-curricular activities. By introducing a computerized system, Victory School can enhance its operational efficiency and ensure that all student clubs are managed effectively, benefiting both the students and the school administration.

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2.2 OVERVIEW OF THE EXISTING SYSTEM

The current system at Victory School for managing club memberships, activities, and finances is primarily manual. Student information, club registrations, and financial transactions are recorded on paper or in spreadsheets, leading to several challenges.

The main issues with the existing system include:

1. **Manual Data Entry**: Student details, including club membership and roles, are manually entered and updated, which increases the chances of errors and inconsistencies.
2. **Error-prone Fee Calculations**: Registration fees for different clubs are calculated manually, which often leads to discrepancies and delays.
3. **Inefficient Management of Club Activities**: Details of club activities, including events and revenue-generating activities, are tracked manually. This results in inefficiencies when managing and reporting on these activities.
4. **Lack of Centralized Data**: The absence of a centralized database makes it difficult to quickly retrieve and manage student and club information, slowing down administrative processes.

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The existing system is cumbersome, error-prone, and lacks efficiency, highlighting the need for an automated system to streamline operations, improve accuracy, and enhance the overall management of the school's clubs.

2.2.1 SYSTEM STRUCTURE

The existing system at Victory School lacks an integrated structure, relying on manual and fragmented processes for managing club memberships, activities, and finances. Each club operates independently, with club member details, activities, and financial records stored separately. This decentralized approach makes it challenging to retrieve relevant information across clubs and hampers the efficiency of club management.

The key issues with the current structure include:

1. **Disparate Data Storage**: Member information, fees, and club activities are stored in different locations (e.g., physical records, spreadsheets), making data retrieval and management difficult.
2. **Lack of Standardization**: Without a unified data model, there is no consistency in how information is recorded across clubs. This results in redundant data entry and potential errors in the system.

3. **Manual Processes**: Each club manages its financial transactions manually, which leads to delays and inaccuracies in fee computations, activity reports, and financial summaries.
4. **Limited Reporting and Data Access**: The lack of an integrated structure hinders the school's ability to generate comprehensive reports, track club revenues, and calculate the school's financial contributions.

The proposed database system is designed to address these structural limitations by centralizing and organizing data, ensuring streamlined club management, accurate fee calculations, and efficient financial tracking.

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2.3 OVERVIEW OF THE PROPOSED SYSTEM

2.3.1 SYSTEM OVERVIEW

The proposed system for Victory School introduces a comprehensive and computerized database solution designed to streamline the management of club memberships, activities, and finances. By replacing manual processes, the system offers a centralized platform for managing student information, club membership details, and financial transactions. The new system integrates all aspects of club operations, ensuring a unified approach to record-keeping, fee calculations, and activity management.

Key features of the proposed system include:

1. **Centralized Data Storage**: The system will store and manage all club-related data in a single location, improving data consistency and accessibility.
2. **Automated Membership Registration**: New students joining clubs will be able to register seamlessly through the system, and their details (including admission number, class, and club memberships) will be captured and stored automatically.
3. **Fee Management**: The system will automatically calculate registration fees based on the club selected, ensuring accurate fee collection and reducing manual errors.
4. **Activity and Event Tracking**: Clubs can log their activities, including revenue-generating events, making it easier to track financial performance and allocate funds accordingly.
5. **Real-Time Reporting**: The system will provide real-time access to financial summaries, club activities, and reports on the school's financial contributions to club events.

With a robust architecture, the proposed system will significantly enhance Victory School's ability to manage club operations efficiently, ensuring accurate financial tracking and facilitating quick decision-making.

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2.3.2 OBJECTIVES OF THE PROPOSED SYSTEM

The proposed system for Victory School is designed to meet several key objectives, directly addressing the challenges faced by the current manual processes:

1. **Centralized Data Management**: Create a centralized repository for club membership data, activities, and financial records, improving consistency, data integrity, and accessibility.
2. **Automated Fee and Service Computations**: Automate fee calculations based on club selection and roles (Regular/Executive), ensuring accurate financial transactions and reducing the possibility of manual errors.
3. **Standardized System Architecture**: Establish a standardized architecture that reduces redundancy, improves data entry processes, and facilitates the seamless cross-referencing of club data.
4. **Real-Time Data Access**: Enable real-time access to critical data, supporting prompt decision-making and better responsiveness to student inquiries and administrative needs.
5. **Comprehensive Reporting**: Provide comprehensive financial reports, including club revenues, expenses, and the school's contributions to annual events, aiding in transparent financial analysis and strategic planning.
6. **User-Friendly Interface**: Design an intuitive and easy-to-use interface for club administrators, teachers, and students, ensuring quick adoption and effective system utilization.
7. **Scalability**: Ensure the system is scalable to accommodate future club expansions, additional activities, and changes in club membership, supporting the long-term needs of Victory School.

The proposed system will address the existing gaps, leading to improved club management, accurate financial tracking, and efficient operations, all contributing to an enhanced experience for students and staff at Victory School.

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2.3.1 BENEFITS OF THE PROPOSED SYSTEM

The implementation of the proposed computerized database system at Victory School is expected to deliver several advantages that will positively impact the efficiency and effectiveness of club management operations. The key benefits include:

1. **Enhanced Operational Efficiency**: Automating membership registration, fee calculations, and activity tracking will reduce manual effort, speeding up processes and increasing overall operational efficiency.
2. **Accuracy in Financial Transactions**: Automated fee and event calculations will minimize the risk of errors, ensuring accurate billing, financial tracking, and transparent invoicing for students.
3. **Centralized Data Accessibility**: A centralized database will provide real-time access to student and club data, improving data availability and reducing response times for both staff and students.
4. **Improved Decision-Making**: With comprehensive and timely reports on financial data, student memberships, and club activities, the system will enable better decision-making and effective management of club operations.
5. **Streamlined Record Management**: The standardized structure of the system will reduce redundant data entry and improve the integrity of student and club records across different activities.
6. **User-Friendly Interface**: The system's intuitive interface will facilitate easy adoption by staff, requiring minimal training for effective use.
7. **Scalability and Adaptability**: Designed to scale with the school's growth, the system will adapt to new clubs, additional students, and changing club activities in future years.
8. **Customer Satisfaction**: The efficient management of memberships and activities, along with accurate financial handling, will enhance student and parent satisfaction, improving the school's overall reputation.
9.

The proposed system will transform Victory School's club management, making processes more efficient, accurate, and user-friendly.

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2.3.2 DISADVANTAGES OF THE PROPOSED SYSTEM

Despite the numerous benefits of the proposed computerized database system, there are several challenges and disadvantages that should be considered:

1. **Initial Implementation Costs**: The transition to a computerized system will incur upfront costs, including software development, hardware upgrades, and staff training, which could strain the school's budget.
2. **Learning Curve**: Staff and students used to manual systems may experience difficulties in adjusting to the new system, potentially causing a temporary dip in productivity during the transition period.
3. **Technical Dependencies**: The system's operation depends on IT infrastructure, and any technical issues such as system failures or internet connectivity problems could hinder accessibility and operations.
4. **Data Security Concerns**: The digital nature of the system raises concerns about the security of sensitive student information. It will be crucial to implement robust cybersecurity measures to protect data from unauthorized access.
5. **System Downtime**: During the system implementation or routine updates, there could be periods of downtime, affecting real-time access to club records and financial data.
6. **Dependency on Skilled Personnel**: The system will require trained personnel to maintain and troubleshoot technical issues. A lack of in-house expertise may necessitate hiring IT support services, adding to ongoing costs.
7. **Resistance to Change**: Staff accustomed to traditional methods may resist using the new system, requiring effective change management strategies to ensure smooth adoption.
8. **Customization Challenges**: The system may need to be customized to meet specific needs as they arise, and this may require ongoing adjustments, which could be time-consuming and costly.

By addressing these potential disadvantages proactively, Victory School can minimize their impact and ensure the successful implementation and long-term benefits of the proposed system.

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2.3.1 SCOPE OF THE SYSTEM

The proposed computerized database system for Victory School's club operations is designed to cover all aspects of club management, including student registration, fee processing, activity tracking, and financial management. The system will include the following key modules:

1. **Student Registration Module**: This module will facilitate the registration of new students into various clubs, capturing essential details such as admission number, name, class (form and stream), and other required information.
2. **Club Management Module**: The system will allow the creation and categorization of clubs, enabling the assignment of patrons and the specification of registration fees for each club. It will also track member roles within the clubs (e.g., Regular/General, Executive/Leaders).
3. **Fee Management Module**: This module will handle the calculation of registration fees, categorize them by club, and ensure accurate payment tracking. It will also manage any additional fees related to club activities.
4. **Club Activities and Events Module**: The system will allow clubs to plan and document their revenue-generating activities, including events such as tournaments or fundraisers. It will track activity details such as name, date, and amount collected.
5. **Financial Transactions Module**: This module will automate the calculation and tracking of the school's contribution to club activities (70% of the allocated amount), as well as managing the club's savings (remaining 30% after the event and operational expenses).
6. **Reporting Module**: The system will feature comprehensive reporting tools to generate summaries of club finances, revenue from activities, contributions, and overall financial performance. It will provide both detailed and summary reports for efficient decision-making.
7. **User Management Module**: This will manage user roles and permissions to ensure that only authorized personnel can access certain features of the system, including financial data and sensitive student information.
8. **System Administration Module**: The system administrator will have the ability to configure and maintain system settings, manage user accounts, and perform updates to keep the system in line with evolving school needs.

While the proposed system aims to streamline and automate many of the processes currently handled manually, it will not completely replace all administrative tasks. Some aspects of student management, decision-making, and club oversight will still require human intervention. The scope of the system is focused on addressing the outlined objectives to ensure effective and efficient club operations at Victory School.

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2.3.2 COST AND BENEFITS ANALYSIS

Cost Analysis:

The implementation of the proposed computerized database system for Victory School involves several cost components, each contributing to the overall financial investment required. The key cost elements include:

1. **Software Development Costs**: This includes expenses related to the design, development, and customization of the system to meet the specific needs of Victory School. It will cover both initial programming and ongoing adjustments.
2. **Hardware Costs**: The school may need to upgrade or acquire new hardware, such as servers, computers, and networking equipment, to support the system's operations and ensure optimal performance.
3. **Training Costs**: Successful system adoption will require investment in training programs for staff and club leaders. This will cover training on the use of the system, entering student and financial data, and troubleshooting any issues that may arise.
4. **Implementation Costs**: This involves the expenses related to setting up and configuring the system, integrating it into the school's operations, and any additional expenses for system testing, data migration, and troubleshooting during the rollout phase.
5. **Maintenance and Support Costs**: Ongoing expenses will include regular system maintenance, updates, and technical support. This will ensure the system remains up-to-date and functional, and any operational issues are promptly addressed.

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Benefits Analysis:

While the upfront costs for the proposed system are substantial, the long-term benefits are expected to provide a significant return on investment, particularly through enhanced operational efficiency and improved financial management. Key benefits include:

- **Time Savings**: Automating administrative tasks will save considerable time for school staff, allowing them to focus on other important activities, such as student engagement and event planning.

- **Accurate Financial Tracking**: With automated fee and revenue management, the school can ensure precise financial tracking, leading to fewer errors and better resource allocation.
- **Improved Reporting**: Detailed financial and activity reports will enable better oversight of club finances, helping to make data-driven decisions and allocate resources more effectively.
- **Student Satisfaction**: Streamlined club registration and management processes, combined with transparent financial handling, will improve student satisfaction and overall engagement in co-curricular activities.

The proposed system is expected to result in better resource management, smoother operations, and increased student involvement in clubs, thus contributing positively to the overall educational experience at Victory School.

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Benefits Analysis for the Proposed Computerized Database System for Victory School Club Membership System

The implementation of the proposed computerized database system for Victory School's club membership management is expected to provide substantial benefits that will enhance the efficiency, transparency, and effectiveness of managing student club activities. The system is designed to automate key processes, ensuring smooth operation of the school's extracurricular activities. The expected benefits include:

1. **Operational Efficiency:** By automating processes such as membership registration, fee collection, role assignment, and event tracking, the system will significantly reduce manual paperwork and administrative tasks. This will streamline operations, saving time and effort, and enabling staff to focus on higher-value tasks such as student engagement and event planning.
2. **Accuracy and Precision:** The system will ensure that all financial transactions, including registration fees and club activity revenues, are recorded accurately. By minimizing the risk of human error in fee calculations and financial reporting, the system will enhance the precision of invoicing, payments, and financial summaries, ensuring transparency and accountability.
3. **Real-time Accessibility:** The centralized nature of the database will allow for immediate access to updated information, including student membership details, role assignments, financial records, and event schedules. This real-time access will empower administrators, patrons, and teachers to make quick, informed decisions and provide prompt responses to student queries.
4. **Informed Decision-Making:** With detailed reporting features, the system will provide administrators with insightful, up-to-date reports on club membership statistics, financial summaries, and event outcomes. These reports will support data-driven decision-making, helping to improve club management and resource allocation while enabling better strategic planning for future club activities.
5. **Improved Record Management:** The database's organized structure will streamline data management, making it easier to store, retrieve, and update club-related records. By eliminating redundancies and enhancing data consistency, the system will support better cross-referencing between different student clubs and activities, improving overall data integrity.
6. **Enhanced Member and Faculty Satisfaction:** With quicker processing of registrations, fees, and membership updates, both students and faculty members will experience a smoother, more responsive system. This ease of use will contribute to higher satisfaction, as students can easily join or exit clubs, and faculty can efficiently manage club activities and financials.

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7. **Scalability and Adaptability:** The proposed system is designed with scalability in mind. As the school grows and more clubs or students join, the system can expand to accommodate additional clubs, students, and activities. This scalability ensures that the system can adapt to the school's evolving needs, from increasing membership to adding new co-curricular offerings.

8.

While the initial setup costs may be significant, the long-term benefits far outweigh these investments. The system will not only improve operational efficiency but also contribute to a more streamlined, organized, and financially transparent management of Victory School's clubs and societies.

Feasibility Study for the Proposed Computerized Database System for Victory School Club Membership System

The feasibility study for implementing the proposed computerized database system at Victory School is a crucial step in assessing the practicality and viability of the project. This study evaluates various factors, including technical, operational, and financial considerations, to ensure the system aligns with the school's objectives and will deliver a positive return on investment.

Through this feasibility analysis, decision-makers will gain a clear understanding of the system's potential benefits and challenges, providing the foundation for making informed decisions throughout the implementation phase. By thoroughly evaluating the system's alignment with the school's needs and goals, the study will guide the planning and execution of the project, ensuring that it delivers the desired outcomes in enhancing the management of student clubs and their activities.

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2.4.1 Economic Feasibility for Victory School Club Membership System

The economic feasibility study evaluates the financial aspects of implementing the proposed computerized database system for Victory School's club membership management. The primary goal is to analyze the costs involved in developing and maintaining the system, alongside the anticipated benefits, to determine if the investment is financially justified.

- **Initial Costs:** The costs involved in system development, including software licensing, customization, and hardware upgrades, are crucial in determining the initial investment. Additionally, expenses for training staff and faculty members to use the system will be accounted for.
- **Ongoing Costs:** The study will also account for ongoing expenses such as system maintenance, regular software updates, and technical support. These costs should be manageable within the school's annual budget.

- **Benefits and Return on Investment (ROI):** The system will streamline processes such as membership registration, fee collection, and event management, which will reduce administrative workload and improve accuracy in financial reporting. These efficiencies will result in long-term cost savings. Additionally, the system will allow for better resource allocation, supporting future growth and more efficient club operations. The analysis will also consider intangible benefits, such as increased satisfaction from both students and faculty, and enhanced organizational transparency, contributing to overall school success.

By comparing the costs with the expected benefits, this study will determine whether the proposed system is financially viable and whether the investment in the system will offer a favorable return over its lifestyle.

2.4.2 Technical Feasibility for Victory School Club Membership System

The technical feasibility study assesses the technical aspects of implementing the proposed computerized database system for managing club memberships at Victory School. This study includes an analysis of the school's existing technological infrastructure to determine its capacity to support the new system, as well as the potential technical challenges.

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- **Infrastructure Assessment:** The school's current network, server capacity, and hardware will be reviewed to ensure they meet the system's requirements. If necessary, upgrades such as additional storage or more powerful servers may be required.
- **Software Requirements:** The system's software needs, such as compatibility with existing operating systems, database software, and potential integration with other school management tools, will be assessed. The goal is to ensure seamless operation and minimal disruption to existing services.
- **Skilled Personnel:** The feasibility study will evaluate the availability of skilled staff or the need for training to manage and maintain the system. The involvement of technical support services or in-house IT staff will be necessary to ensure smooth operation post-implementation.
- **Integration Challenges:** Any technical difficulties associated with data migration from existing systems or the integration of various functionalities

(e.g., student records, financial transactions) will be identified and addressed.

This analysis ensures that the proposed system can be integrated into the current technological environment without significant hurdles.

2.4.3 Operational Feasibility for Victory School Club Membership System

Operational feasibility evaluates how well the proposed system will fit into the day-to-day operations of Victory School, considering the impact on workflows, staff adaptation, and potential disruptions during the transition from manual to automated processes.

- **Impact on School Operations:** The system will automate various processes such as membership registration, club fee management, and event tracking, which will significantly streamline administrative workflows. This will free up time for staff to focus on other strategic activities.
- **Ease of Adoption:** The study will evaluate how easy it will be for staff and faculty to adopt the system. This includes assessing user-friendliness, the learning curve, and the required training. Proper training programs will be planned to ensure smooth transition and effective use of the system.
- **Transitioning to Automation:** Transitioning from a manual system to a computerized system could cause some temporary disruption. The feasibility study will assess the potential challenges during this shift, including resistance to change and the need for temporary workaround solutions to maintain club activities without disruption.
- **Efficiency Gains:** The system is expected to improve the efficiency of day-to-day operations by automating repetitive tasks and reducing errors. By eliminating the need for paper-based tracking and manual calculations, the system will provide real-time data, improving overall operational efficiency.

2.4.4 Legal Feasibility for Victory School Club Membership System

Legal feasibility examines the proposed system's adherence to relevant laws, regulations, and standards, ensuring compliance with data protection laws, privacy regulations, and any other legal considerations that may apply.

- **Data Protection and Privacy:** The system must comply with the relevant data protection laws and privacy regulations, such as Kenya's Data Protection Act (2019). The study will ensure that personal data, such as student information, is stored and processed in a secure and legally compliant manner, with appropriate consent mechanisms for collecting and using data.

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- **Confidentiality of Student Records:** Measures to protect sensitive student data and maintain confidentiality will be considered. This includes access controls, data encryption, and regular audits to safeguard personal and financial information from unauthorized access.
- **Compliance with Educational Standards:** The system will also be evaluated to ensure it aligns with the Ministry of Education's regulations for managing student and club data, particularly regarding transparency in financial reporting and student engagement.

This study will ensure that the system complies with all legal requirements, reducing the risk of legal issues for the school.

2.4.5 Schedule Feasibility for Victory School Club Membership System

Schedule feasibility evaluates the timeline required for the development, testing, and implementation of the proposed system, ensuring it can be completed within the expected timeframe without compromising quality.

- **Project Milestones:** A clear timeline will be created, detailing key milestones such as requirements gathering, system design, software development, testing, training, and final deployment. Realistic deadlines will be set for each phase to ensure timely completion.
- **Potential Delays:** The feasibility study will identify potential risks and delays, such as technical challenges or delays in staff training. Contingency plans will be put in place to address these issues promptly to minimize impact on the project timeline.
- **Alignment with School Year:** The project timeline will be aligned with Victory School's academic calendar to avoid disruption during peak periods, such as the start of a school year or final exams. The system should be ready for full implementation before significant club activities begin.

The study ensures that the proposed system can be delivered on time, within budget, and without compromising on quality, helping the school to achieve its operational goals efficiently.

The comprehensive feasibility study for the proposed system will provide a holistic evaluation of its viability, covering the economic, technical, operational, legal, and schedule-related aspects. This thorough analysis will guide Victory School in making an informed decision about whether to proceed with the implementation of the computerized database system for club membership management.

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2.3 Fact Finding for Victory School Club Membership System

Fact-finding is a critical phase in system analysis that involves gathering information about the current processes, user needs, and overall requirements of the proposed system. The goal of this phase is to understand the existing processes within Victory School's club management, identify areas for improvement, and capture the specific needs of the users. Several fact-finding methods were used to ensure a comprehensive understanding of the current operations and system requirements for the new computerized database system.

5.1 Interviews

Structured interviews were conducted with key stakeholders, including school administrators, faculty members, club leaders, and end-users (students). These interviews were designed to gather insights into their roles and responsibilities, as well as to capture their thoughts on the existing club membership management processes. Discussions focused on identifying pain points in the manual registration, fee collection, and event tracking processes. Key feedback from faculty and staff highlighted areas for improvement in administrative efficiency and the need for better communication tools for club activities.

5.2 Document Review

A thorough review of existing documents was conducted, including student records, current club membership databases, fee structures, event schedules, and communication logs. This method helped to understand the flow of information, data structures, and existing business rules governing club operations at the school. Document review provided insight into the gaps in the current system, such as the lack of integration between student records and club membership data, leading to duplicated efforts in managing both sets of information.

5.3 Observation

Observational techniques were employed to gain a firsthand understanding of the daily operations involved in managing club memberships at Victory School. This included observing how staff and students interact with the current manual system for membership registration, fee payment, and event participation. The aim was to identify operational bottlenecks, such as delays in data entry, inconsistencies in record-keeping, and inefficiencies in tracking club activities. Additionally, observations helped highlight the challenges faced by club leaders in managing memberships and communicating with students.

5.4 Questionnaires

A set of questionnaires was distributed to a representative sample of students, faculty, and administrative staff to collect quantitative data and opinions regarding their experiences with the current club membership system. The questionnaire covered various aspects of the current system, including ease of use, reliability,

satisfaction with data accuracy, and suggestions for improvement. The responses provided valuable insights into user preferences, challenges faced with the existing system, and specific features that should be incorporated into the proposed computerized database system.

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5.5 Prototyping

Prototyping was used to create initial versions of the proposed system, providing a tangible representation of the system's potential interface and functionality. This iterative approach allowed stakeholders to interact with the prototype, offering feedback on user interface design, data input processes, and overall usability. The prototyping process helped in validating design assumptions and ensuring that the system met the actual needs and expectations of users. It also facilitated early detection of any usability issues, ensuring that the final system would be user-friendly and intuitive.

2.4 Summary of Fact-Finding Methods Used

The fact-finding phase for the Victory School club membership system employed a diverse set of methods to gather comprehensive information about current operations and system requirements. Interviews were essential in extracting detailed insights from key stakeholders, including administrators, faculty, and students, about their roles and challenges. Document review provided an in-depth understanding of the current system's data structure, flow, and limitations. Observation techniques allowed the project team to identify inefficiencies and bottlenecks in real-time operations. Questionnaires provided quantitative data and user feedback, offering valuable insights into system usability and areas for improvement. Prototyping provided a visual representation of the proposed system, facilitating user feedback and design validation.

VICTORY SCHOOL CLUB System

The combination of these fact-finding methods ensured a well-rounded understanding of the current system’s strengths and weaknesses, guiding the development of a targeted and effective computerized database system that meets the operational needs of Victory School’s club membership management.

2.3 System Requirements and Specifications for Victory School Club Management System

The system requirements and specifications for the proposed computerized database system for Victory School's club management are outlined below to ensure that the system is designed to meet both the technological and operational needs of the school. These specifications will guide the development process, ensuring the system is capable of handling the school’s club registration, fee management, and event tracking.

Hardware Requirements:

- **Processor:** Intel Core i5 or equivalent
- **RAM:** 8GB

- **Hard Disk:** 1TB
- **Monitor:** 21-inch display (or equivalent)
- **Keyboard and Mouse:** Standard peripherals suitable for use with the system

VICTORY SCHOOL CLUB
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Software Requirements:

- **Operating System:** Windows 10 or later (or equivalent)
- **Database Management System:** Microsoft SQL Server or equivalent (for managing relational data)
- **Application Software:** Custom-built or off-the-shelf software application with a graphical user interface (GUI) that interfaces with the database to handle membership, fee collection, and event management tasks.

Functional Requirements:

1. **Student Registration Module:**
    - The system should allow for the seamless registration of new students in various clubs. It should capture essential details such as student name, grade, contact information, club preferences, and emergency contact information.
    - The system should support the management of existing memberships, allowing students to renew or update their details as needed.
2. **Club Management Module:**
    - The system should allow for easy management of club information, such as club names, descriptions, meeting times, and assigned faculty members.
    - It should support categorizing clubs by types (e.g., sports, academic, cultural, etc.) and facilitate easy creation, deletion, or modification of clubs.
3. **Fee Management and Payment Module:**
    - The system should automatically calculate and track the fees associated with each club, including any discounts or scholarships.
    - It should generate invoices for students and provide mechanisms for fee payment tracking.
    - Payment history should be easily accessible for both students and administrative staff.

VICTORY SCHOOL CLUB System

4. **Event Management Module:**

    - The system should allow for event scheduling, tracking, and registration, including information on event dates, locations, registration deadlines, and participant lists.
    - The system should notify students about upcoming events and allow students to register for events directly through the platform.

5. **Reporting Module:**

    - The system should include robust reporting features, allowing the generation of reports related to club memberships, event participation, fee payments, and overall school-wide club activities.
    - Reports should be exportable in various formats such as Excel or PDF for administrative use and decision-making.

VICTORY SCHOOL CLUB
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6. **User Management Module:**
    - The system should support role-based access control, ensuring that different users (administrators, faculty, students) have appropriate permissions to access and manage club data.
    - Faculty members should be able to manage their respective clubs, while administrative staff should have full access to all club-related data.

7. **System Administration Module:**
    - The system should include administrative controls to manage overall system settings, user access, and system updates.
    - Administrative users should be able to configure the system as needed based on the changing needs of the school and its clubs.

Non-Functional Requirements:

1. **Performance:**
    - The system should provide efficient response times, even during peak usage periods (such as when students are registering for clubs or events).
    - The system should be optimized for handling multiple concurrent users without significant delays or performance degradation.

VICTORY SCHOOL CLUB System

2. **Reliability:**
    - The system should be highly reliable, with minimal downtime and the ability to recover quickly in the event of a failure.
    - Data integrity and consistency should be maintained, ensuring no loss of student or financial data.

3. **Security:**
    - The system should implement robust security features, including encryption of sensitive data (e.g., student contact information, payment details).
    - The system should ensure that unauthorized access to student or financial data is prevented through secure login mechanisms and data protection measures.

4. **Scalability:**

   - The system should be designed to scale as the school grows, with the ability to accommodate an increasing number of students, clubs, events, and transactions without significant rework.
   - The system should also be able to handle additional features or modules if needed in the future.

VICTORY SCHOOL CLUB
System

5. **Usability:**
    - The system should feature an intuitive, user-friendly interface that is easy for both administrative staff and students to navigate.
    - Clear instructions, tooltips, and a simple design should make the system accessible to users with varying levels of technical expertise.

6. **Compatibility:**
    - The system should be compatible with a range of devices (desktops, laptops, tablets) and web browsers (Google Chrome, Mozilla Firefox, Safari).
    - It should be responsive to ensure compatibility with various screen sizes and devices, ensuring usability across multiple platforms.

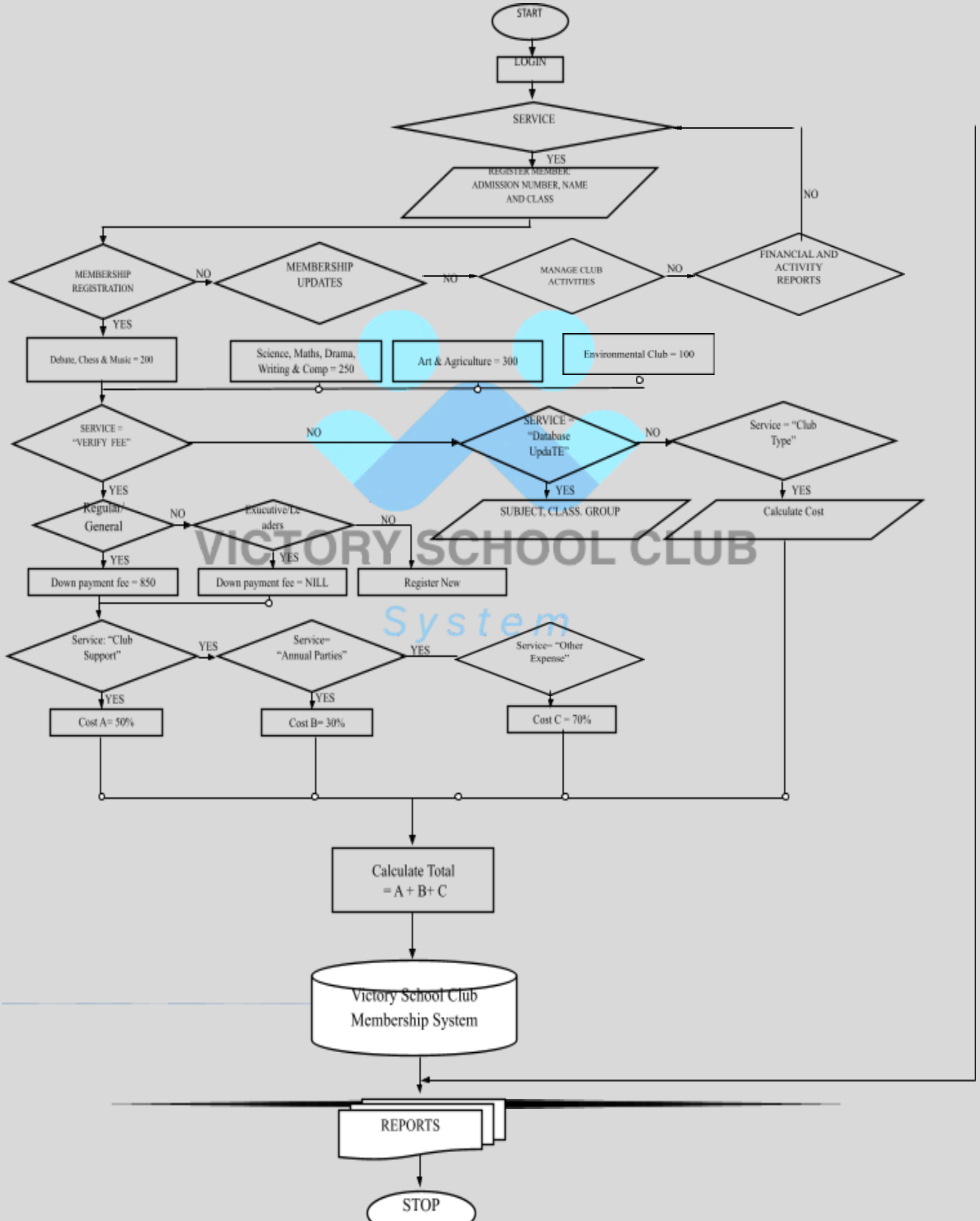
These hardware, software, functional, and non-functional requirements form the foundation for the design and development of the proposed system, ensuring that it meets Victory School's specific needs for managing club memberships, events, and financial transactions. By adhering to these specifications, the system will be positioned to improve efficiency, accuracy, and overall management of school clubs, benefiting both students and administrative staff.

VICTORY SCHOOL CLUB
System

CHAPTER 3: SYSTEM DESIGN

3.1 PRELIMINARY DESIGN

3.1.1 Overall System Flowchart

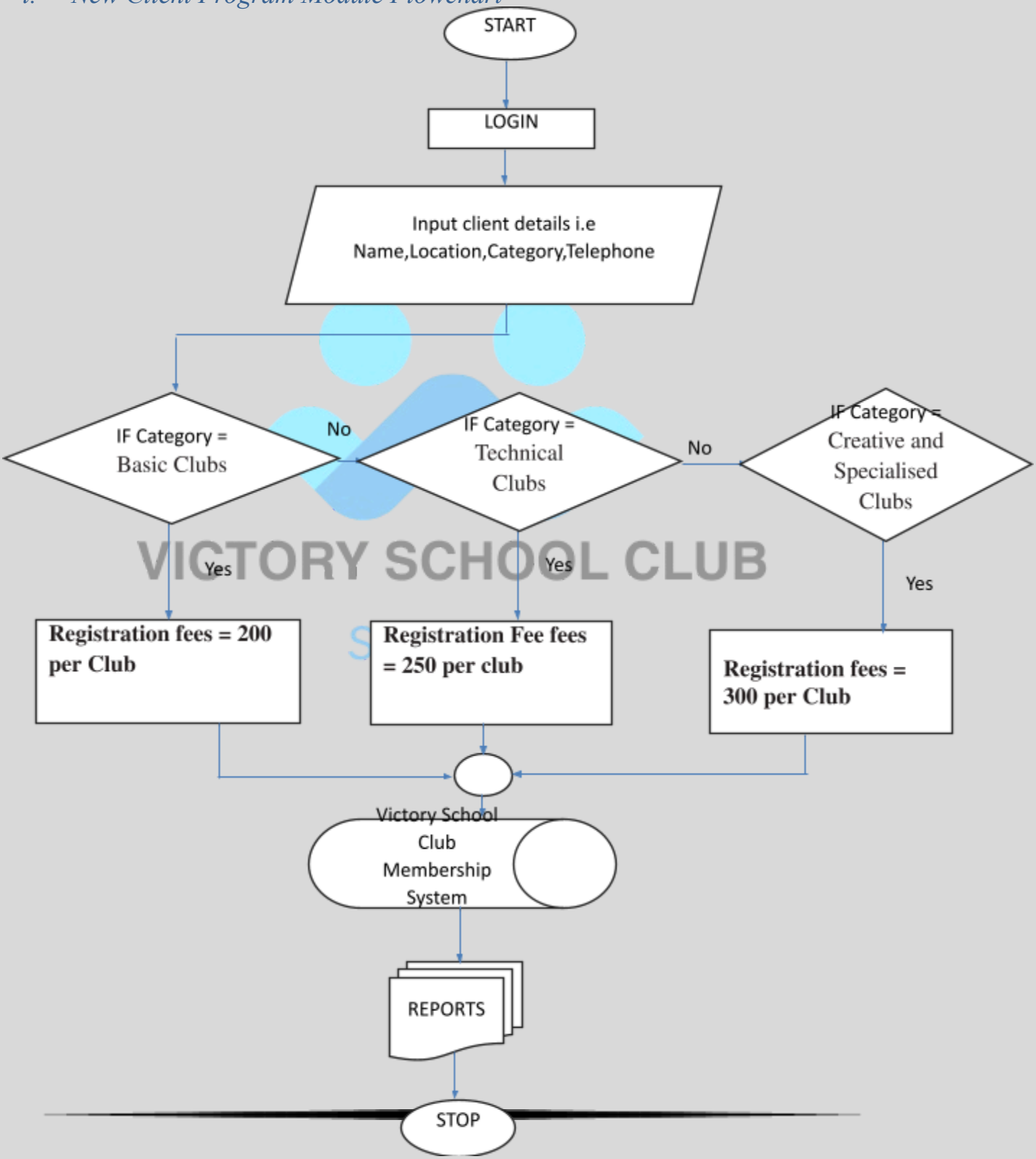


VICTORY SCHOOL CLUB System

3.1.2 PROGRAM MODULE FLOWCHARTS

These flowcharts describe the process of modules that are run individually inside the whole system during normal operations. They include;

i. New Client Program Module Flowchart



VICTORY SCHOOL CLUB

3.2 Detailed Design for Victory School Club Management System

The detailed design phase outlines the key tables necessary for the efficient operation of the proposed system. Below is the design for the database structure, which includes essential fields that support functionalities like student registrations, club management, fee tracking, event management, and more.

3.2.1 Tables

Students Table

Field Name	Data Type	Description
StudentID	AutoNumber	Unique identifier for each student.
FirstName	Text	First name of the student.
LastName	Text	Last name of the student.
Grade	Text	Grade or year level of the student.
DateOfBirth	Date/Time	Date of birth of the student.
ContactNumber	Text	Primary contact number of the student.
EmailAddress	Text	Email address of the student.
EmergencyContact	Text	Emergency contact details for the student.
ClubID	Number	Foreign key referring to the club the student is registered in.

Clubs Table

Field Name	Data Type	Description

ClubID	AutoNumber	Unique identifier for each club.
ClubName	Text	Name of the club (e.g., Soccer, Drama, Science, etc.).
ClubType	Text	Type of the club (e.g., Academic, Cultural, Sports, etc.).
FacultyAdvisor	Text	Name of the faculty member supervising the club.
MeetingDays	Text	Days of the week when the club meets.
MeetingTime	Text	Time when the club meetings occur.
ClubFee	Currency	Fee associated with the club.

Event Management Table

Field Name	Data Type	Description
EventID	AutoNumber	Unique identifier for each event.
EventName	Text	Name of the event (e.g., Club Fair, Sports Day, etc.).
EventDate	Date/Time	Date when the event is scheduled.
EventLocation	Text	Location of the event.
EventDescription	Text	Detailed description of the event.
ClubID	Number	Foreign key referring to the club organizing the event.
RegistrationDeadline	Date/Time	Date by which students need to register for the event.

Fee Payment Table

Field Name	Data Type	Description

PaymentID	AutoNumber	Unique identifier for each payment.
StudentID	Number	Foreign key referencing the student who made the payment.
AmountPaid	Currency	Amount paid for club membership or event participation.
PaymentDate	Date/Time	Date when the payment was made.
PaymentMethod	Text	Method of payment (e.g., Cash, Credit Card, Bank Transfer).
TransactionReference	Text	Reference number or ID associated with the payment.

Tax and Fee Structure Table

Field Name	Data Type	Description
FeeStructureID	AutoNumber	Unique identifier for each fee structure.
ClubFee	Currency	The base fee for club membership.
EventFee	Currency	Fee associated with event registration.
Discount Percentage	Decimal	Percentage discount applied to club or event fees.
TaxRate	Decimal	Percentage representing the tax rate applicable to fees.

Table 3.1 Sample Tables Used and Field Properties

These tables define the core structure for organizing data for student registrations, club membership, event management, fee payments, and the overall financial transactions within the system.

The relationships between the tables will be defined as follows:

- **Student-Club Relationship:** A many-to-one relationship where each student can be associated with one club, but a club can have multiple students.
- **Club-Event Relationship:** A one-to-many relationship where each club can host multiple events.
- **Student-Payment Relationship:** A one-to-many relationship where each student can make multiple payments (for club fees or event participation).
- **Fee Structure Application:** The Fee and Tax Rates will be used in conjunction with student and event payments to calculate final amounts, taking into account possible discounts.

This database structure is designed to support the core functionalities required by Victory School's club management system, with the flexibility to expand as the school's club offerings and event management needs grow.

Additional Considerations: Further normalization and indexing may be applied to ensure optimal performance and scalability. Relationships and foreign key constraints will be used to maintain referential integrity across the system, ensuring that data is consistent and reliable.

VICTORY SCHOOL CLUB
System

3.2.1 INPUT DESIGN (FORMS)

3.2.2.1 New Student Form

Fees

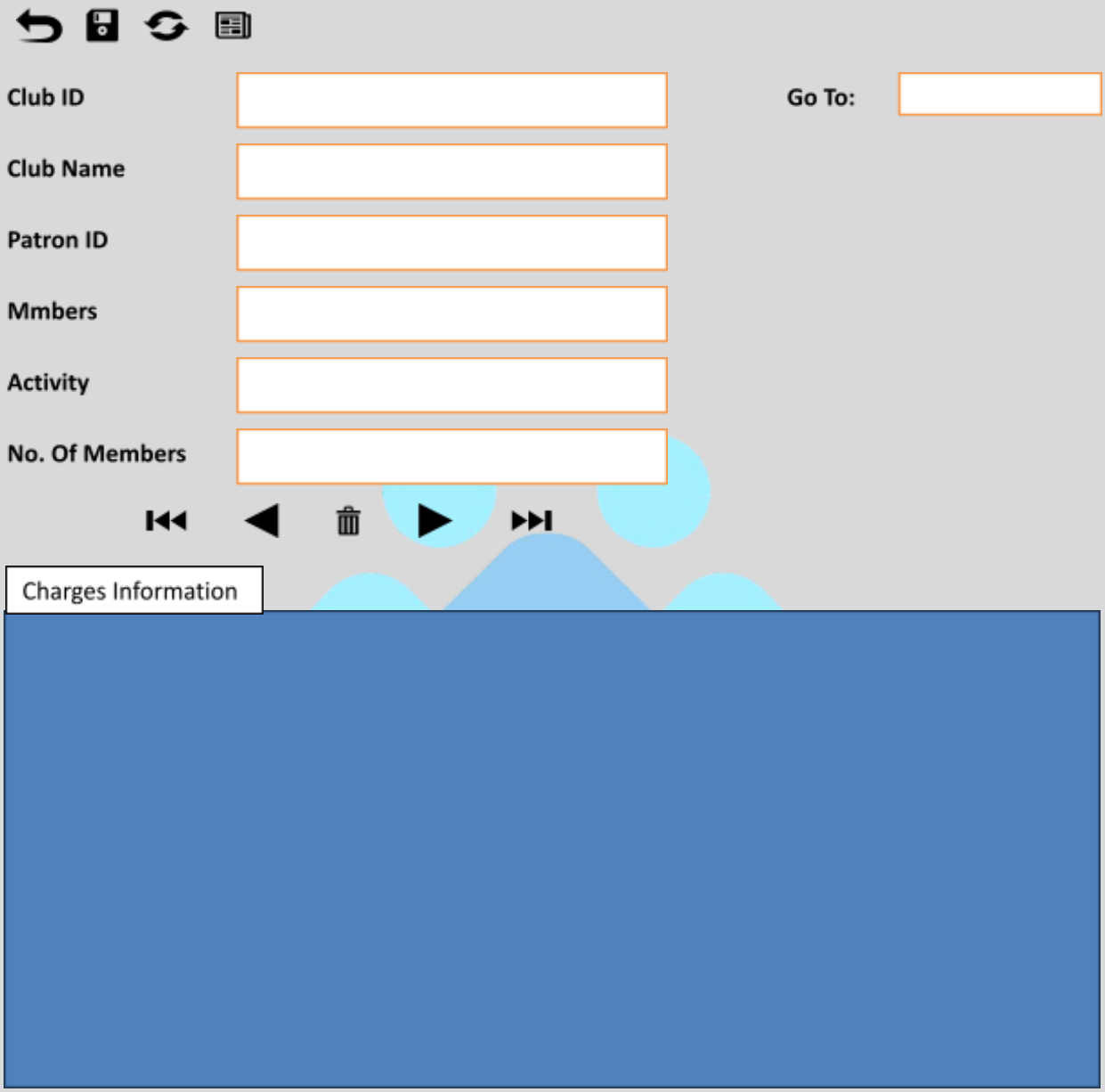
Club Type

Client ID	Name	Club Type

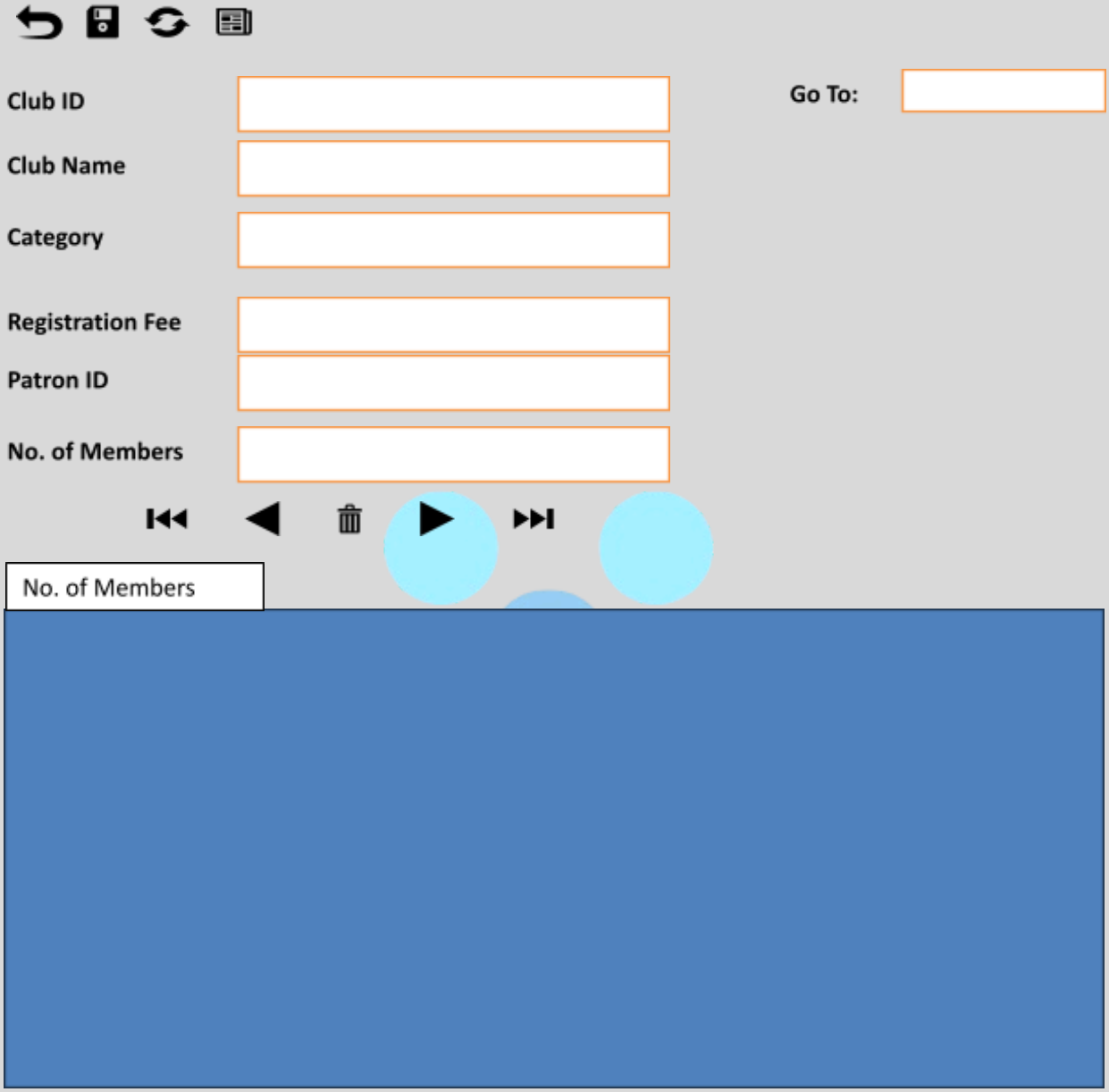
VICTORY SCHOOL CLUB

System

3.2.2.2 Club Activities Form



3.2.2.3 New Club Form



⏮	- Navigates to the top most record
◀	- Navigate to the latter record
🗑	- Removes the selected record from the system's database
▶	- Navigate to the subsequent record
⏭	- Navigate to the most bottom record

VICTORY SCHOOL CLUB
System

3.2.2 OUTPUT DESIGN (REPORTS)

3.2.3.1 Student's Report

Victory School Club Membership System

Students Report

StudentID	First Name	Adm. Number	Class	Club Joined
		Total Fees Paid		

3.2.3.2 Club List Report

Victory School Club Membership System

Clubs Table

ClubsID	Club Name	Category	Reg. Fee	No. of Members

3.2.3.3 Activity Report

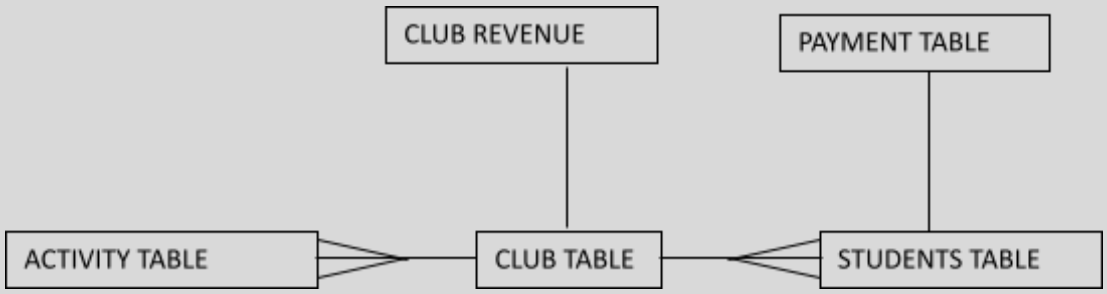
Victory School Club Membership System

ACTIVITIES TABLE

ActivityID ID	Club ID	Activity Name	Revenue	Description
		Grand Totals		

3.2.4 RELATIONSHIPS

Relationship Diagram



VICTORY SCHOOL CLUB
System

3.1 GENERAL SYSTEM SECURITY

Once the computerized system has been created, it's important to ensure that it is secure to protect the data and information that it stores. Here are the general system security measures that were implemented:

1. User authentication: The system should require users to authenticate themselves using a username and password to prevent unauthorized access.
2. Access control: Access to different parts of the system should be restricted based on user roles and privileges. For example, only authorized users should be able to view or modify sensitive data.
3. Data encryption: Sensitive data, such as user passwords or financial information, should be encrypted to prevent unauthorized access.
4. Regular backups: Regular backups should be taken to ensure that data can be recovered in the event of a system failure or data loss.
5. Physical security: Physical access to the system should be restricted to authorized personnel only, and the system should be kept in a secure location.
6. Security updates: The system should be kept up to date with the latest security updates and patches to address any known vulnerabilities.
7. Monitoring and logging: The system should have logging and monitoring capabilities to track user activity and detect any suspicious behavior.
8. Disaster recovery: A disaster recovery plan should be in place to ensure that the system can be quickly restored in the event of a catastrophic event.

By implementing these security measures, this ensured that the computerized system is secure and protected the data and information it stores.

VICTORY SCHOOL CLUB System

CHAPTER 4:

SYSTEM CONSTRUCTION

In this chapter, we will implement the Victory School Club Membership System in Microsoft Access. The system will be designed to efficiently handle the various activities associated with managing club memberships, activities, finances, and reports. This system will be constructed using the .accdb format in Microsoft Access, ensuring compatibility with versions from Access 2010 to the latest Office 365

DATABASE RELATIONSHIP DIAGRAM

The database relationship diagram is an implementation of the design shown earlier.

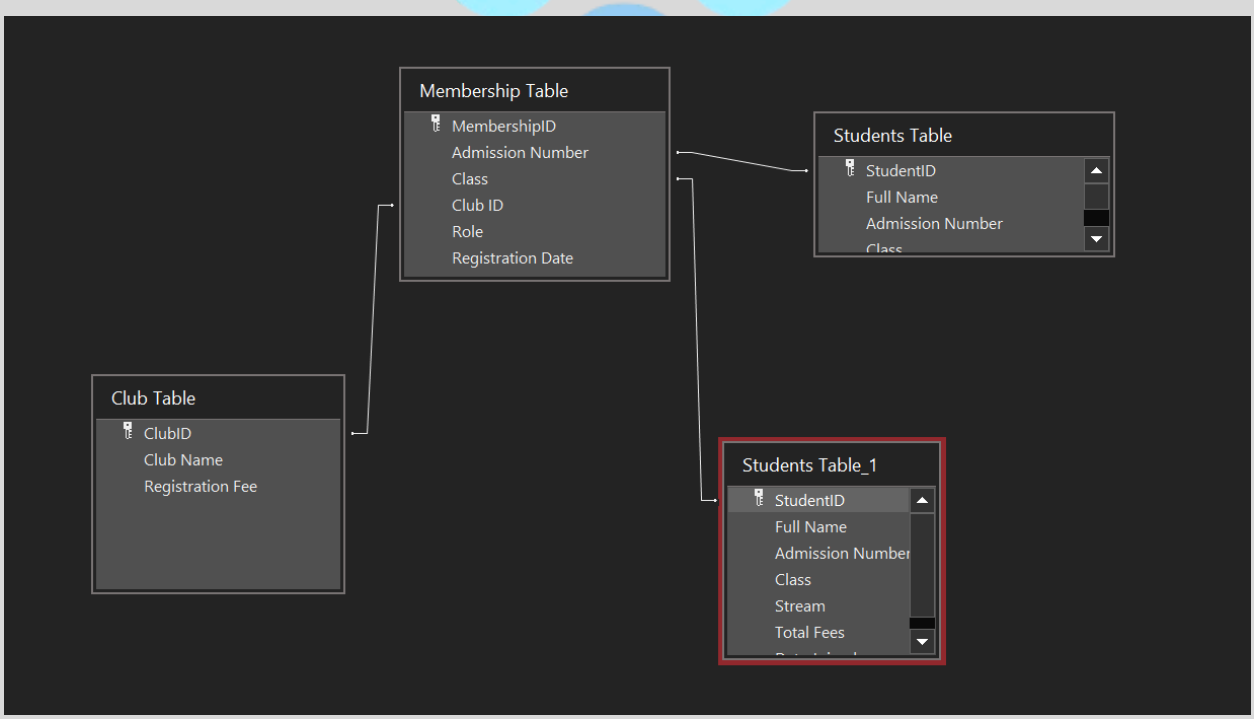


Figure 4.1 Database Relationship

4.1 ENFORCING INTERGRITY CONSTRAINTS

Ensuring data integrity is essential to maintain accuracy and consistency in the Victory School Club Membership System. This chapter highlights the methods applied to enforce integrity constraints, including:

- **Entity Integrity**: Ensures each record in the database is uniquely identifiable.
- **Referential Integrity**: Guarantees relationships between tables are maintained accurately.
- **Domain Integrity**: Ensures data entered into fields adheres to specified formats and constraints.
- **User-Defined Constraints**: Custom rules set to enforce specific business logic for managing club memberships, finances, and activities.

4.2.1 REFERENCIAL INTERGRITY

Referential integrity ensures that a record entered in the related (secondary) table has a matching primary key in the primary table. This is achieved by using a foreign key that create the relationship. Figure 4.2 shows referential integrity between the Clients and Drilling Applications tables.

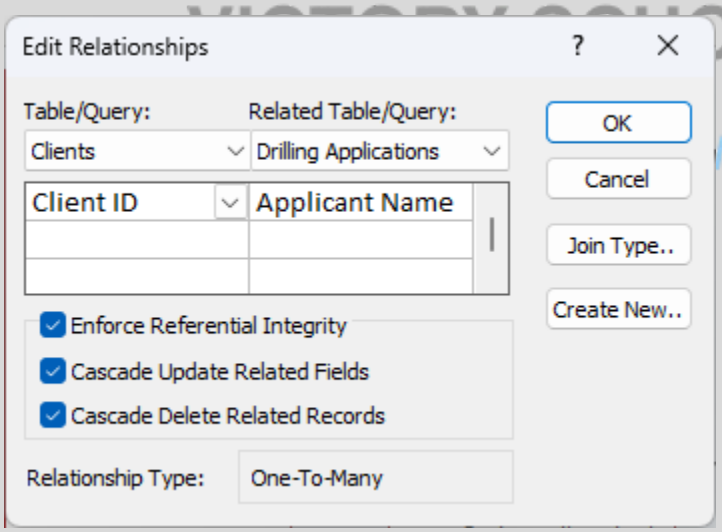


Figure 4.2 Referential Integrity

4.2 Database Tables

4.3.1 STUDENTS TABLE

The Students's table shown in design view below is used to store details of the Company's clients.

StudentID	Full Name	Admission Numb	Class	Stream	Total Fees	Date Joined
001	Linus Abdikadir	Jane Onyango	2N	0723530125	Ksh. 20,000.00	1/17/2025
002	Njuguna Yegon	Immanuel Oloo	1E	0789656842	Ksh. 50,000.00	1/22/2025
003	Fidel Wanje	Ian Kiama	3N	0779663146	Ksh. 24,000.00	1/16/2025
004	Ramadhan Owiti	Jane Onyango	4W	0754810947	Ksh. 252,000.00	1/29/2025
093					Ksh. 0.00	
(New)					Ksh. 0.00	
Total	**4**					

Figure 4.3 Student's Table

VICTORY SCHOOL CLUB
System

4.3.2 PATRON TABLE;

Patron Table

PatronID	Patron Name	Contact Num	Email	Club ID	Role Descrip
001	Jane Onyango	0711347783	janeon@gmail.com	Debate Club	
002	Immanuel Oloo	0731133222	Immanuelo@gmail.com	Science Club	
003	Ian Kiama	0731213224	IanKiama@gmail.com	Writing Club	
(New)					

Figure 4.4: Patron's Table

4.3.3 Membership Table

Membership Table

Membershi	Admission Number	Class	Club ID	Role	Registration
001	Ramadhan Owiti	1E	Music Club (Bar	Regular/Genera	1/2/2025
002	Fidel Wanje	4W	Mathematics	Exucutive Leade	1/9/2025
003	Linus Abdikadir	3N	Writing Club	Regular/Genera	1/8/2025
(New)					

Figure 4.5: Membership Table

4.3.4 Club Finance

Membership Table | Club Finance

FinanceID	Club ID	Total Revenue	Last Update
1	Music Club (Band)	Ksh. 30,000.00	1/16/2025
(New)		Ksh. 0.00	

Figure 4.6:Club Finance

4.3 DATABASE QUERIES

In order to combine several tables, we need to create queries that would be used to process various transactions and generate important reports

VICTORY SCHOOL CLUB System

4.4.1 Membership Query

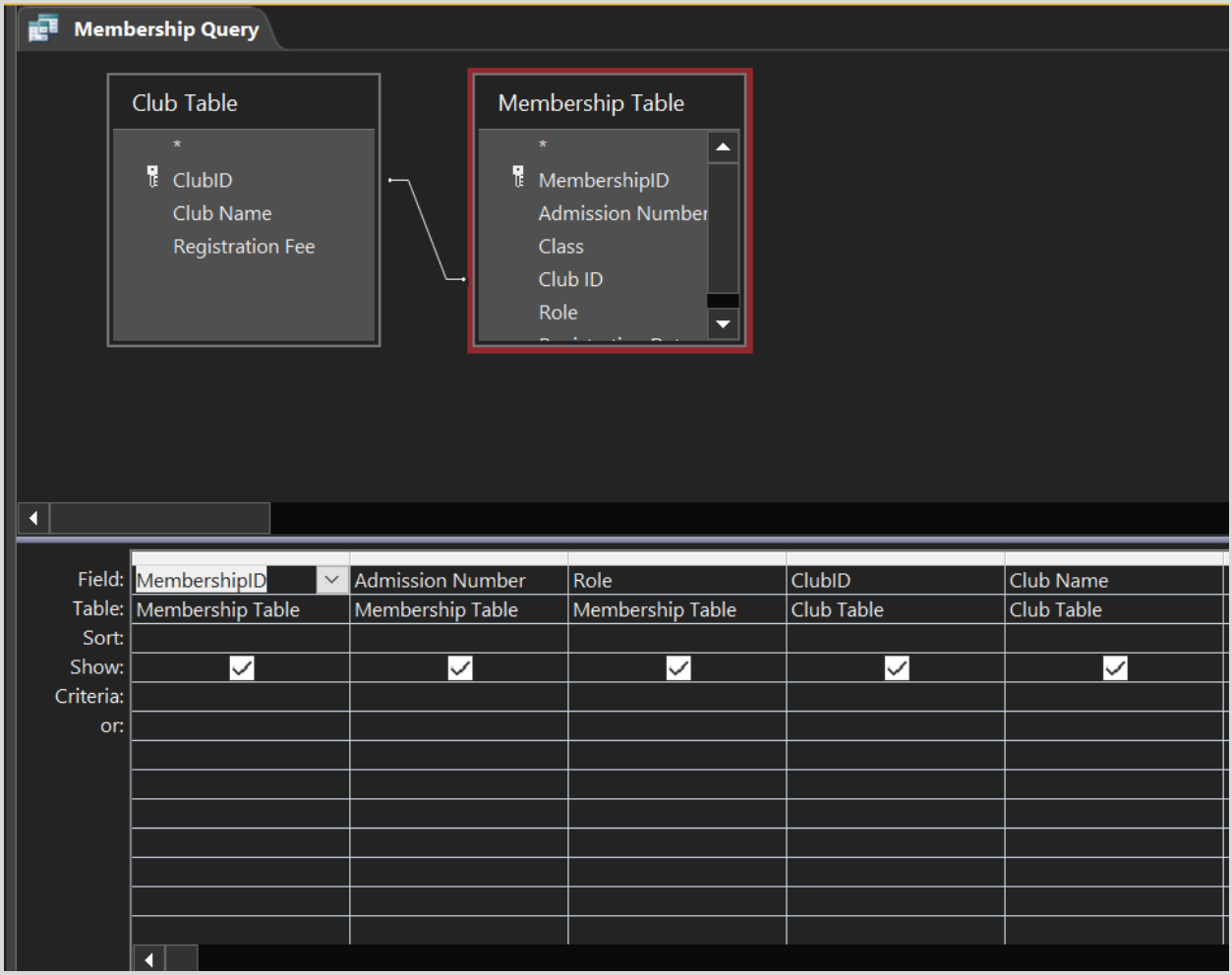


Figure 4.5 Membership Query

System

4.4.1 Club Table Query

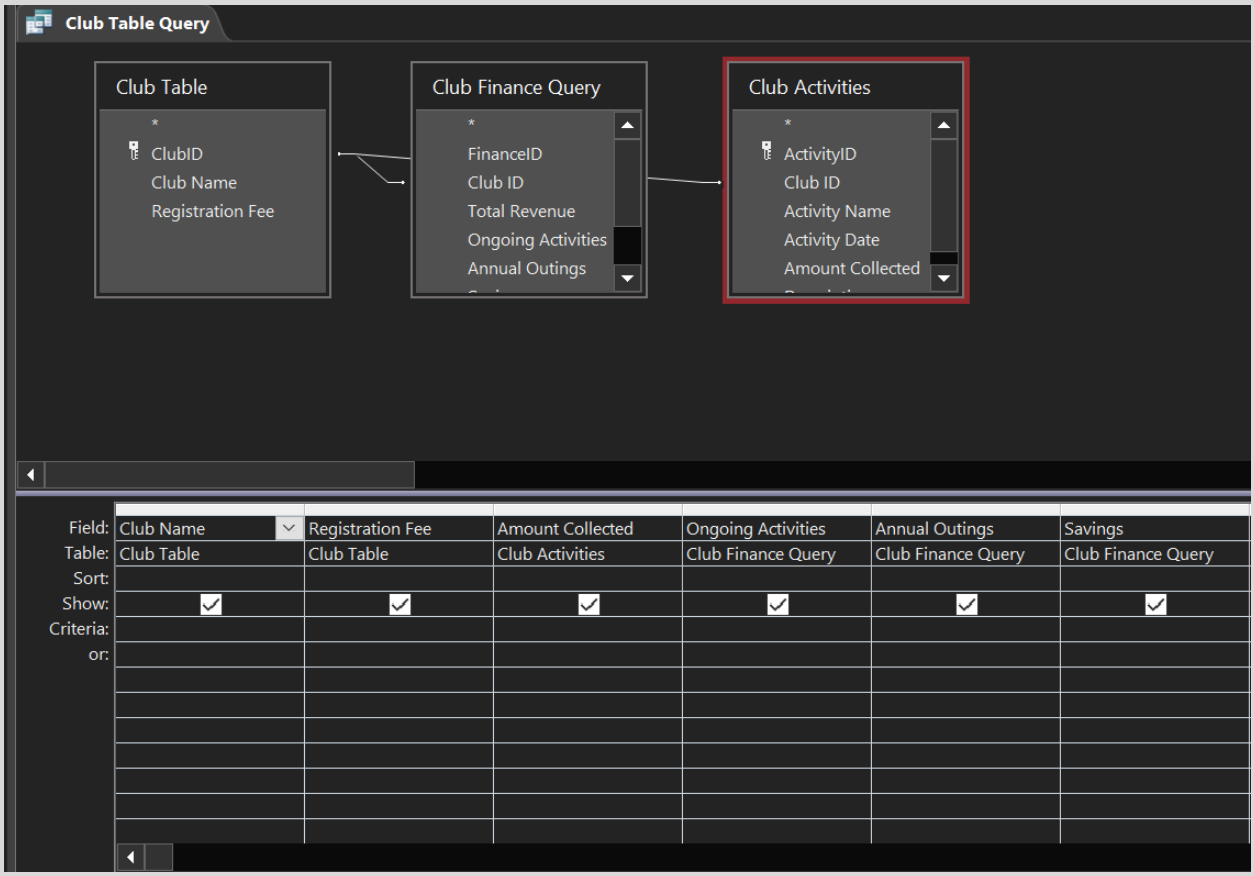


Figure 4.6 Club Table Query

VICTORY SCHOOL CLUB System

CHAPTER 5: USER MANUAL

5.1INTRODUCTION

This User Manual is designed to guide you through the various features and functionalities of the system. Whether you're a student, a club patron, or a school administrator, this manual will help you navigate the system with ease and ensure that you can efficiently manage club memberships, activities, and finances.

From student registration to generating financial reports, the system offers a range of tools to streamline the management of school clubs. In this manual, you'll find step-by-step instructions, explanations, and visuals to help you perform tasks effectively. The goal is to equip you with the knowledge to make the most out of the system for smooth and efficient club operations.

5.1.1 Software Requirements

i. GUI based operating system
ii. Microsoft Access application software from office 2010 or later.
iii. Antivirus software

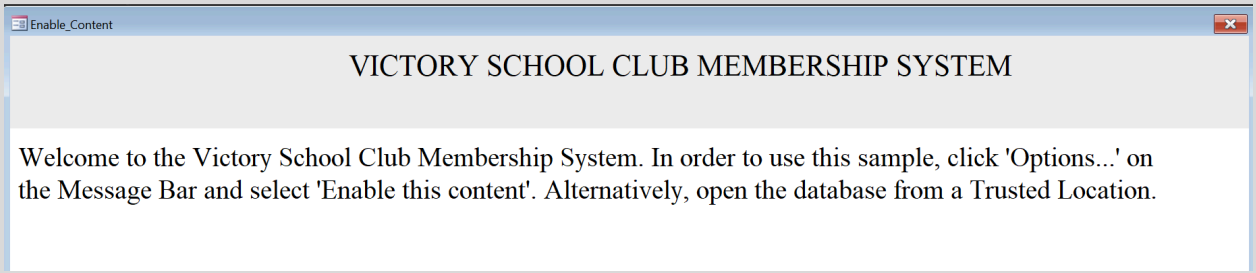
5.1.2 Hardware requirements

i. A computerized system
ii. 488-66MHZ processor
iii. At least 170MB of RAM
iv. A compact disk since the software comes on a compact disk
v. Printers
vi. POS terminal equipment

VICTORY SCHOOL CLUB System

5.2 HOW TO INSTALL, START AND RUN THE SYSTEM INTERFACE

The system software is copied from the installation disk and pasted in the documents folder. A shortcut icon may be created in the in the desktop. Confirm the presence of a **Microsoft access Software** in the machine. Now click on the icon to launch the system. After opening the system, a screen appears that prompts the user to enable content for first time use. Click on **Enable Content** to terminate.



A Startup Screen will appear in which the Program will load its components for approximately 5 seconds. Kindly be patient till the program fully boots as illustrated.

Figure 5.2 Start-up Screen

Figure 5.1: Login Screen

A User log in screen will appear as shown Below:

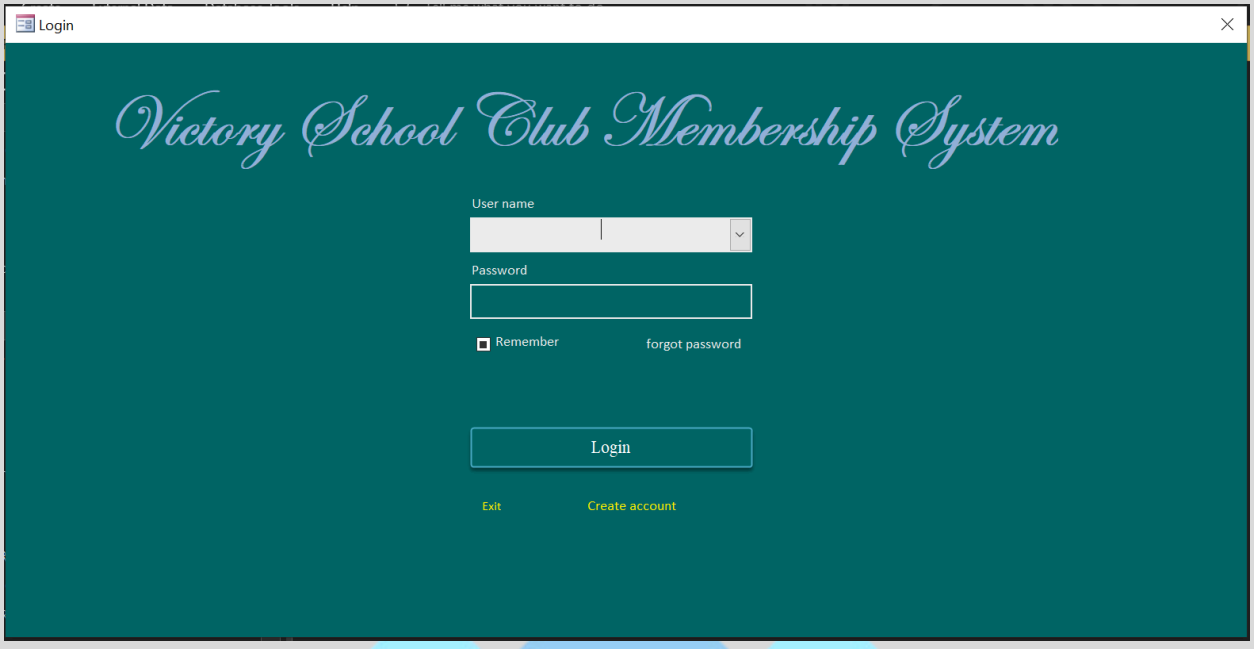


Figure 5.1: Login Screen

VICTORY SCHOOL CLUB
System

5.3DATA MANIPULATION

During data manipulation, a main home screen is used. This is very user friendly and easy to use. The Home screen looks as shown below. Click on any desired option to launch the required operation.

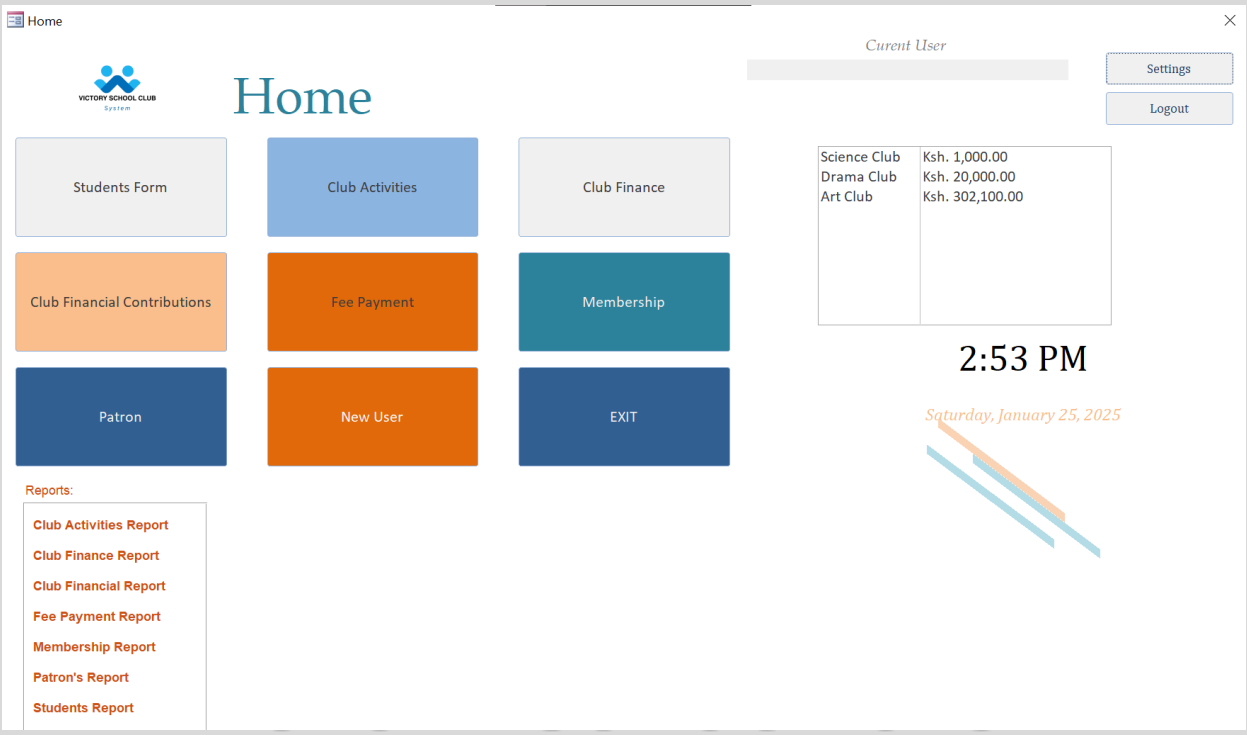
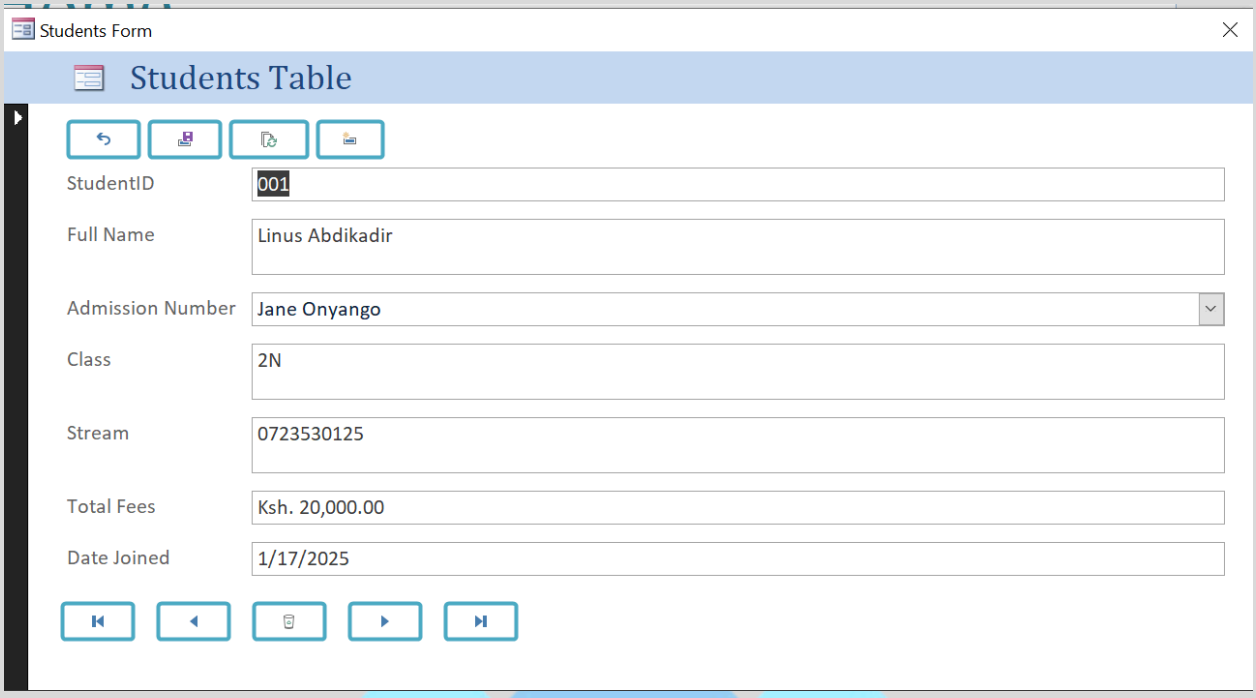


Figure 5.1 Main Switchboard

5.4NEW STUDENT

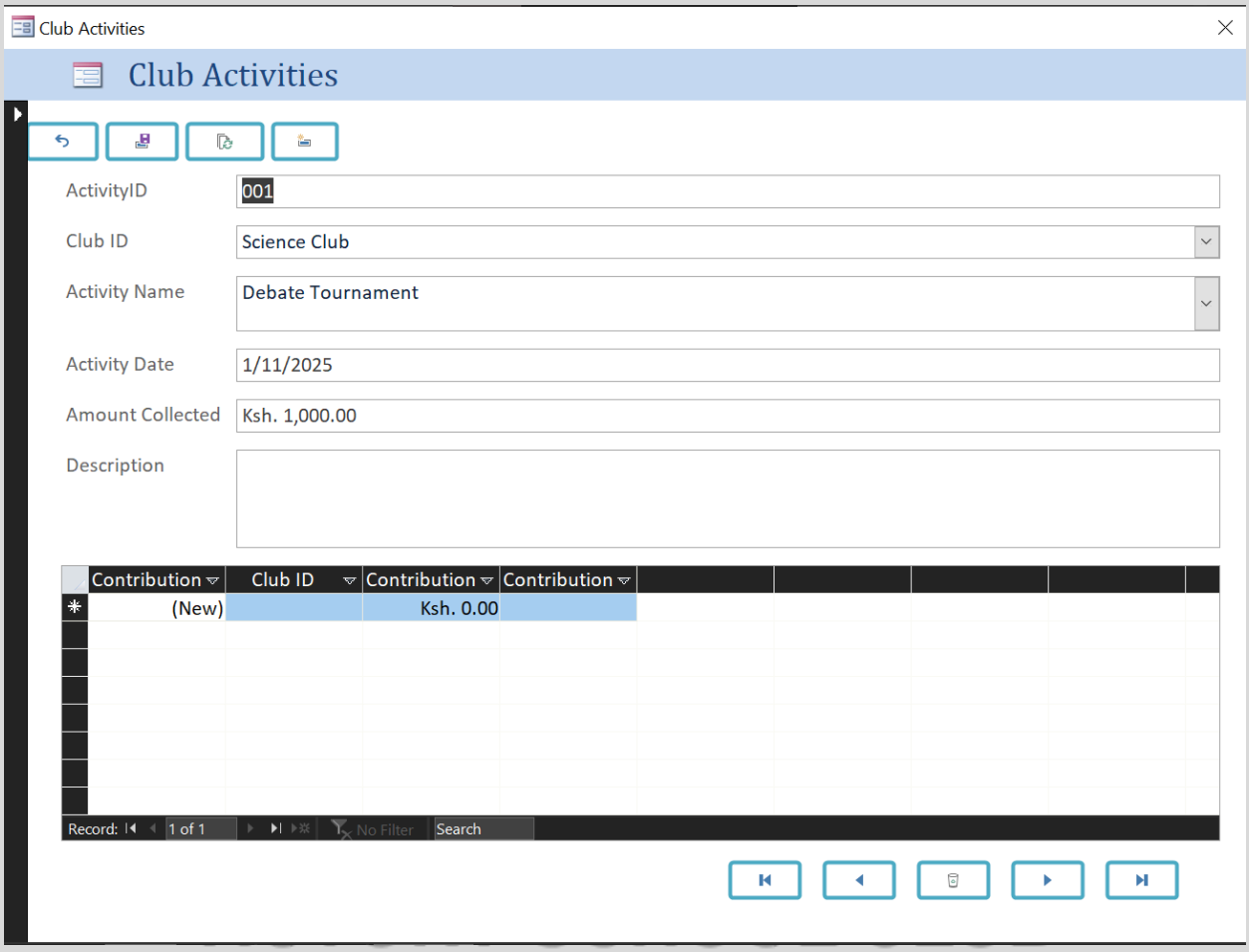
5.4.1 New Client



5.4.2 New StudentApplication

VICTORY SCHOOL CLUB
System

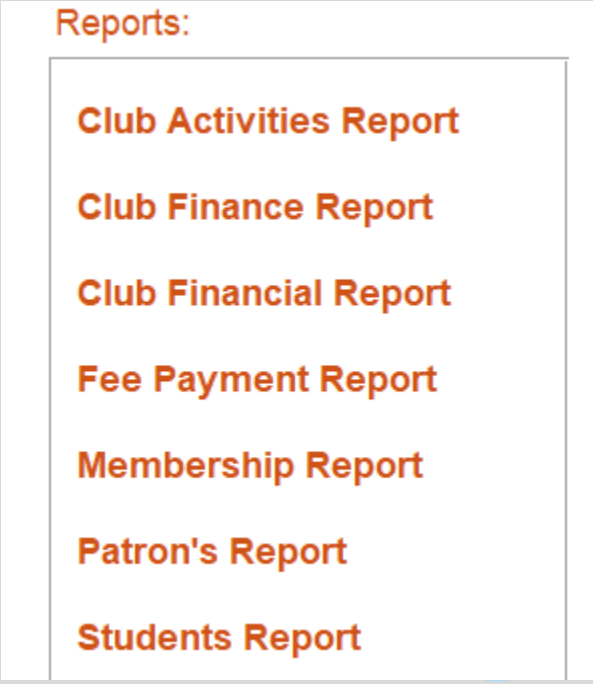
5.4.3 Club Activities



System

5.5GENERATING REPORTS

To generate group reports, click on the buttons aligned on the right of the main switchboard as shown below:



VICTORY SCHOOL CLUB
System

Sample of existing Reports are as shown

5.5.1 Student Report

Students Report

Students Table

Saturday, January 25, 2025
2:59:49 PM

StudentID	Full Name	Admission Number	Class	Stream	Total Fees	Date Joined
001	Linus Abdikadir	Jane Onyango	2N	0723530125	Ksh. 20,000.00	1/17/2025
002	Njuguna Yegon	Immanuel Oloo	1E	0789656842	Ksh. 50,000.00	1/22/2025
003	Fidel Wanje	Ian Kiama	3N	0779663146	Ksh. 24,000.00	1/16/2025
004	Ramadhan Owiti	Jane Onyango	4W	0754810947	Ksh. 252,000.00	1/29/2025
093					Ksh. 0.00	
					Ksh. 346,000.00	

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VICTORY SCHOOL CLUB
System

5.5.2 Patron Table Form

Patron's Report

Patron Table

Saturday, January 25, 2025
3:01:19 PM

PatronID	Patron Name	Contact Number	Email	Club ID	Role Description
001	Jane Onyango	0711347783	janeon@gmail.com	Debate Club	
002	Immanuel Oloo	0731133222	Immanuelo@gmail.com	Science Club	
003	Ian Kiama	0731213224	IanKiama@gmail.com	Writing Club	

3

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5.5.3 Membership Report

Membership Report

Membership Table

Saturday, January 25, 2025
3:02:04 PM

MembershipID	Admission Number	Class	Club ID	Role	Registration Date
001	Ramadhan Owiti	1E	Music Club (Band)	Regular/General Member	1/2/2025
002	Fidel Wanje	4W	Mathematics	Exucutive Leader	1/9/2025
003	Linus Abdikadir	3N	Writing Club	Regular/General Member	1/8/2025

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5.5.4 Fee Payment Report

Fee Payment Report

Fee Payment

Saturday, January 25, 2025
3:02:40 PM

PaymentID	StudentID	ClubID	Amount	Payment Date	Payment Status
1	Linus Abdikadir	Music Club (Band)	Mathematics	1/2/2025	Pending
1					

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5.6ADVANCED FEATURES OF THE SYSTEM

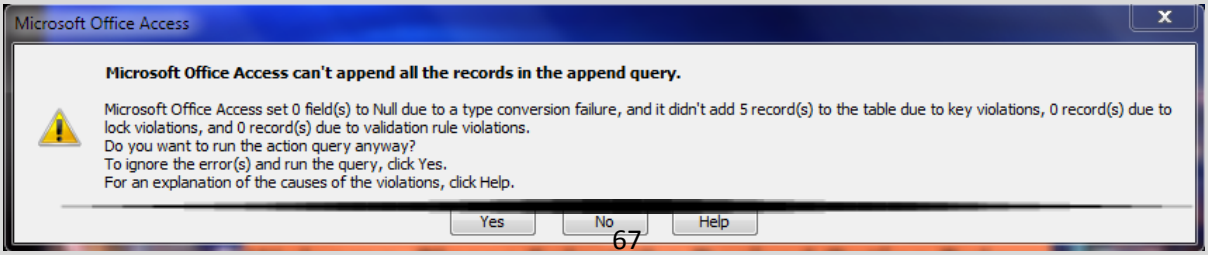
The Log In list button shows the details of every user log in and time of log in which is able to track any malicious activities.

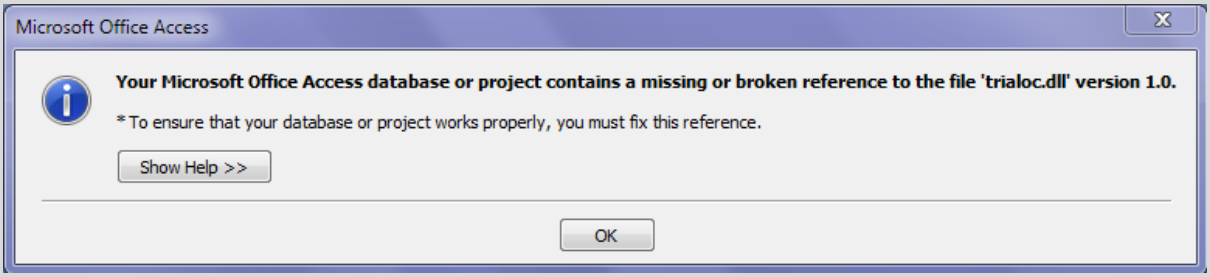
logTable

logID	userName	tDate	tTime
86	Admin, System	7/13/2023	8:46:59 PM
87	Admin, System	2/11/2024	5:09:43 PM
88	SDS	2/11/2024	6:12:22 PM
89	SDS	2/11/2024	6:12:58 PM
90	SDS	2/11/2024	6:18:53 PM
91	name, testing	2/11/2024	6:23:51 PM
92	name, testing	2/11/2024	6:28:37 PM
93	name, testing	2/11/2024	7:45:24 PM
94	name, testing	2/11/2024	7:50:04 PM
95	name, testing	2/12/2024	12:39:50 AM
96	name, testing	2/12/2024	12:40:27 AM
97	SDS	2/12/2024	1:03:32 AM
98	name, testing	2/12/2024	2:31:32 AM
99	SDS	2/12/2024	1:56:30 PM
100	SDS	2/13/2024	6:45:45 AM
101	name, testing	2/13/2024	6:45:56 AM
102	SDS	2/13/2024	6:46:28 AM
103	My Admin	2/13/2024	6:53:29 AM
104	My Admin	2/13/2024	6:53:51 AM
105	SDS	2/13/2024	6:54:26 AM

5.7TROUBLESHOOTING GUIDE

Some small and less harmful errors may occur during the use of the system. They have been carefully tested over a period of time and have been found to cause no harmful damage to the data kept in the system. They can be easily bypassed by clicking OK or YES to accept the permissions.





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CHAPTER 6: SYSTEM IMPLIMENTATION

6.1STAFF TRAINING

Staff Training Plan for Victory School Club Membership System

Staff training is essential for the successful implementation of the **Victory School Club Membership System**. This program aims to ensure that all staff members can effectively utilize the system, manage student club memberships, track finances, and generate reports. With proper training, staff will be able to seamlessly transition to using the new system, ensuring smooth operations and maximum productivity.

Training Objectives:

1. **Familiarize staff with the system interface and navigation** – Ensure users can easily access and navigate through the system.
2. **Train staff on data entry procedures** – Teach how to accurately register students, manage club memberships, and record activities.
3. **Educate staff on financial management** – Provide knowledge on how to input financial data, compute registration fees, track club revenue, and allocate funds.
4. **Guide staff on report generation** – Enable staff to generate financial reports and membership summaries for clubs.
5. **Provide troubleshooting techniques** – Train staff to troubleshoot common issues and provide basic support.

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Training Sessions:

The training will be conducted in a classroom-style environment with a blend of theory and practical sessions. Each training session will focus on interactive learning, including hands-on exercises, to ensure a clear understanding of the system.

Training Content:

1. **System Navigation:**

    1. Introduction to the user interface.
    2. Understanding the menu structure and system controls.
    3. Navigating between student data, club information, activities, and financial records.

2. **Data Entry Procedures:**

1. Step-by-step instructions for adding student details (name, class, club membership).
2. Recording club activities and the funds raised.
3. Updating student membership status (e.g., when students join or leave clubs).

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Financial Management:

1. Understanding club registration fees and how they are tracked.
2. Recording and allocating club funds (50% for activities, 30% for parties).
3. Calculating the school's contribution and tracking savings.

Report Generation:

1. Demonstrating how to generate various reports: financial summaries, club membership reports, and activity breakdowns.
2. Customizing and filtering reports based on club, class, and year.

Training Materials:

- **Training Manuals and Guides:** Detailed step-by-step instructions for system use.
- **Presentation Slides:** Visual aids covering key concepts and workflows.
- **Practice Datasets:** Hands-on datasets for staff to practice entering information and generating reports.
- **Troubleshooting FAQs and Guides:** A reference document to help staff resolve common issues.

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Assessment and Feedback:

At the end of each session, staff will be assessed on their understanding of the system. They will complete practical exercises and a short quiz. Feedback will be collected from all participants to understand their challenges and improve future training.

Ongoing Support:

After training, staff will have access to ongoing support through:

- **Helpdesk Support:** To assist with technical issues and questions.
- **Additional Training Sessions:** To address any gaps in knowledge or to provide refresher courses.
- **Regular System Updates:** Notifications about updates to the system and how they may affect daily operations.

By investing in this comprehensive training program, Victory School ensures its staff are proficient in using the Club Membership System, leading to improved management of clubs, efficient financial tracking, and better overall operations.

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6.2 Change-Over Strategy for Victory School Club Membership System

The change-over strategy outlines the approach to transitioning from the current manual processes to the new computerized database system for Victory School's club membership management. A carefully planned change-over strategy is essential to minimize disruption, ensure smooth adoption, and maximize the benefits of the new system.

Change-Over Strategy Used: Parallel Adoption

The **parallel adoption** strategy will be implemented during the transition to the new computerized system. This strategy involves operating both the old manual system and the new system at the same time for a specific period, allowing for a gradual transition while minimizing risks.

Rationale for Parallel Adoption:

1. **Risk Mitigation:** Running both systems concurrently allows for data comparison and validation between the old and new systems, ensuring data accuracy and reducing the risk of loss or inconsistency during the transition phase.
2. **Training and Familiarization:** Staff will have the chance to gradually adapt to the new system while continuing to use the old processes. This reduces resistance to change and eases the adoption of the new system.
3. **Fallback Option:** In case of any issues or challenges with the new system, the old system will remain operational as a backup, ensuring the continuity of club operations without major disruptions.
4.

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Implementation Steps:

1. **System Configuration and Testing:** Ensure the new computerized database system is correctly configured and thoroughly tested. This will confirm that the system meets all functional requirements and is compatible with existing processes.
2. **Parallel Run Phase:** Both the old manual system and the new system will run simultaneously. During this phase, staff will be instructed to enter data into both systems to validate the consistency and accuracy of information.
3. **Monitoring and Evaluation:** Continuous monitoring of the new system will be conducted to gather feedback from staff and identify any issues or areas

for improvement. This feedback will inform necessary adjustments and refinements to the system.

4. **Transition Planning:** Once the new system is fully tested, staff are trained, and data integrity is ensured, the final transition will take place. This may involve migrating data from the old system, conducting final training sessions, and decommissioning the old system.

Communication Plan:

Clear communication will be vital throughout the change-over process. Regular updates will be provided to all stakeholders, including staff and students, to keep them informed of the progress. Training sessions will be scheduled, and open channels for feedback will be established to address any concerns or issues during the transition.

By employing the parallel adoption strategy, Victory School ensures a smooth and seamless transition to the new system, enabling the efficient management of club memberships, activities, and finances with minimal disruption.

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CHAPTER 7: SYSTEM REVIEW

7.1 Objective Appraisal for Victory School Club Membership System

The objective appraisal evaluates the performance, effectiveness, and alignment of the **Victory School Club Membership System** with its predefined objectives. This appraisal, conducted after the implementation, aims to identify the system's strengths, weaknesses, opportunities, and threats to ensure continuous improvement and successful operation.

Evaluation Criteria:

1. **Functionality:** Assessing the system's ability to handle essential tasks such as student registration, club membership management, activity tracking, and financial record keeping accurately and efficiently.
2. **User Experience:** Evaluating the user interface design, ease of navigation, and overall satisfaction to determine how user-friendly the system is for students, staff, and administrators.
3. **Data Integrity:** Verifying the accuracy, consistency, and reliability of data stored in the system, including student membership details, club activities, financial records, and reports.
4. **Performance:** Analyzing how well the system performs in terms of speed, responsiveness, and scalability to ensure it can handle a growing volume of data and users effectively.
5. **Adaptability:** Evaluating the system's flexibility and ability to evolve with future updates, changes, and technological advancements to meet the school's needs.

Findings:

Functionality:

he system effectively manages student registration, club memberships, and financial transactions (fees and revenue from activities). Users report that tasks such as club registration, financial management, and reporting are completed efficiently and accurately.

User Experience:

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Users (staff and administrators) provided positive feedback regarding the user interface and navigation. The system's design is intuitive, and the staff training conducted during implementation contributed to a smooth transition and high user adoption.

Data Integrity:

The system successfully ensured data integrity, with regular checks in place to verify the accuracy and consistency of student records, club memberships, and financial data. No major issues with data corruption or loss were reported during the appraisal.

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Performance:

The system performed well during day-to-day operations, with acceptable speed and responsiveness. However, as more students and clubs are added, further optimization may be necessary to maintain optimal performance as data volume increases.

Adaptability:

The system is flexible enough to accommodate changes, such as adding new clubs or modifying membership roles. While it is adaptable to current needs, future scalability considerations and potential integration with other school systems should be planned for long-term sustainability.

Recommendations:

Performance Optimization:

Conduct a performance analysis to identify potential bottlenecks and implement optimization measures. This will ensure the system maintains its responsiveness as the school grows and additional data is processed.

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Continuous Training:

Ongoing training and support for staff should be prioritized to ensure continued user proficiency. As the system evolves with updates and new features, training will help maintain high adoption rates and effective use.

Enhanced Reporting:

Further enhance the reporting features to provide more detailed insights into club activities, finances, and student participation. Adding advanced filtering options could support more nuanced decision-making.

Future Planning:

Develop a long-term roadmap for the system, considering possible future enhancements, system upgrades, and potential integration with other school

management systems. This will ensure the system remains adaptable and relevant as the school's needs evolve.

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Conclusion:

The objective appraisal of the **Victory School Club Membership System** indicates that the system is performing well and meeting its intended goals. Staff and administrators are satisfied with its functionality, and the system demonstrates strong data integrity and user experience. To ensure long-term success, continued monitoring, optimization, and training will be essential to maintaining a smooth, efficient, and effective club management system at Victory School.

7.1 Challenges Encountered During System Development for Victory School Club Membership System

The development of the computerized database system for **Victory School's Club Membership** was a complex and challenging process. Several obstacles emerged during the development, which required careful planning, problem-solving, and collaboration to overcome. Recognizing these challenges was vital in ensuring the system's successful implementation and long-term success.

Challenges Encountered:

1. **Complexity of School Processes:** Victory School's club membership system needed to manage various processes, such as student registration, club selection, role assignment (executive/regular), fee tracking, and event management. The integration of these processes into a cohesive system posed a challenge, requiring careful mapping and testing to ensure smooth interactions between the system components.
2. **Data Validation and Integrity:** Ensuring the accuracy and integrity of the data input into the system was a significant challenge. The system had to manage a variety of data types, such as student details, club memberships, payment histories, and activity logs. Developing effective data validation rules to prevent errors, inconsistencies, and unauthorized changes required extensive testing and refinement to ensure data reliability.
3. **Fee Calculations and Revenue Management:** The computation of registration fees and revenue distribution among clubs involved multiple calculations based on the type of club and event revenues. Designing algorithms that accurately calculated fees based on club categories, generated revenue, and appropriately allocated funds for the club's activities, annual parties, and savings was a complex task during system development.

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4. **User Interface Design:** Designing an intuitive, user-friendly interface was critical for ensuring staff and students could easily interact with the system. Striking the right balance between functionality, simplicity, and aesthetics posed challenges, particularly when considering the varied technical skill levels of users. Iterative design, prototyping, and user feedback were necessary to refine the interface for better usability.
5. **Integration with Existing School Systems:** Integrating the new system with existing school management software, such as student information systems or financial management tools, was a challenge. Ensuring seamless data exchange between these systems and the club membership system required careful planning, testing, and coordination to prevent data discrepancies or disruptions in school operations.
6. **Resource Constraints:** Development efforts were constrained by limited resources such as time, budget, and skilled personnel. Managing these constraints required prioritization of tasks and efficient resource allocation. Some features had to be scaled back or postponed for later phases of the project to ensure the core functionality was delivered on time.
7. **Testing and Quality Assurance:** The system required thorough testing to ensure its functionality and reliability. Developing comprehensive test cases to simulate real-world scenarios, conducting bug fixes, and addressing inconsistencies in club registrations, payment processing, and report generation demanded significant time and attention. Rigorous testing was necessary to ensure the system was error-free before it went live.

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Despite these challenges, the development team's dedication, coupled with effective collaboration with school stakeholders, allowed the successful implementation of the **Victory School Club Membership System**. Documenting the challenges encountered and how they were overcome provides valuable insights that can help with future updates or similar projects in the future.

7.1 Strengths of the Automated System for Victory School Club Membership System

The computerized database system developed for **Victory School's Club Membership** exhibits several key strengths that contribute to its efficiency, effectiveness, and overall value to the school. These strengths not only optimize

operational processes but also ensure a smooth and sustainable implementation across school activities.

Strengths:

1. **Streamlined Processes:** The system efficiently automates various school activities, including student registration, club membership tracking, fee management, and event planning. By digitizing and centralizing these processes, the system reduces manual paperwork and operational delays, leading to improved administrative efficiency.
2. **Enhanced Accuracy and Consistency:** Automation ensures accurate data entry, fee calculation, and reporting. Built-in validation checks help prevent human errors, ensuring reliable data throughout the system. This consistency is crucial for tracking memberships, payments, and student activities without inconsistencies or discrepancies.
3. **Improved Decision-Making:** With real-time data access, the system provides teachers and administrators with actionable insights into club memberships, financial statuses, and student involvement. This access enhances decision-making, from planning school events to allocating resources for club activities.
4. **Increased Productivity:** By automating routine tasks like fee processing and club registrations, administrative staff can focus on higher-value tasks, such as organizing events or mentoring students. This leads to an overall increase in productivity across the school.

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5. **Scalability and Flexibility:** The system's design accommodates the school's growth and changes in club offerings. It can handle an increase in the number of clubs, students, or financial transactions, ensuring the system remains adaptable to future needs and technological advancements.
6. **Enhanced Student Engagement:** By centralizing student data, including their club memberships and roles, the system makes it easier to monitor student participation, provide tailored support, and engage students in club activities. Teachers and staff can better track student involvement and offer personalized guidance.
7. **Financial Transparency and Accountability:** With detailed tracking of club revenues, registration fees, and financial allocations, the system ensures transparency and accountability. It helps manage finances effectively, ensuring that funds are distributed according to the predefined guidelines and providing the school with an audit trail for financial oversight.

7.2 Weaknesses of the Automated System for Victory School Club Membership System

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While the system offers numerous benefits, certain weaknesses may require attention to enhance its overall functionality and efficiency. Addressing these weaknesses is key to ensuring the long-term success of the system and improving user experience.

Weaknesses:

1. **Initial Implementation Costs:** The development and implementation of the system involve significant upfront costs, including software development, hardware purchases, staff training, and system customization. These costs may pose a financial burden, especially for schools with limited budgets.
2. **Learning Curve for Users:** The shift from a manual or legacy system to an automated platform may present challenges for staff unfamiliar with technology. There may be an initial learning curve for teachers, administrators, and students, leading to temporary inefficiencies and slower adoption.

3. **System Downtime and Technical Issues:** As with any technological system, the possibility of system downtimes or technical issues such as server malfunctions, software bugs, or data synchronization problems exists. Such disruptions may temporarily affect club operations, registration processes, or financial tracking.
4. **Data Security Risks:** Centralizing sensitive student and financial data in the system raises potential security concerns, such as unauthorized access or data breaches. Proper data security measures, including encryption, access control, and regular security audits, are essential to safeguard against these risks.
5. **Integration Challenges:** The new system may face challenges when integrating with existing school management systems or third-party tools, such as payment gateways or event management software. Compatibility issues or data exchange problems may arise, requiring additional technical effort to resolve.
6. **Maintenance and Upkeep:** Continuous system maintenance, including updates, bug fixes, and hardware upkeep, is necessary to ensure smooth operation. The school must allocate sufficient resources to support ongoing system management, which can become a challenge if resources are limited.
7. **Scalability Limitations:** Although the system is designed to scale, there may be limitations in handling an exponentially growing number of students, clubs, or events. As the school expands, the system architecture may need to be reassessed to ensure it can handle larger datasets without performance degradation.

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Identifying and addressing the weaknesses in the **Victory School Club Membership System** is essential to optimize its functionality and ensure its long-term success. By proactively tackling these weaknesses, such as addressing the initial implementation costs, training needs, and security concerns, the school can maximize the benefits derived from the system. With continuous improvements and resource allocation, the system can become a vital tool in enhancing student engagement, administrative efficiency, and overall school operations.

7.1RECOMMENDATIONS

Based on the findings from the review of the automated system for Victory School's club membership management, the following recommendations are proposed to optimize its performance, enhance user satisfaction, and address areas for improvement:

1. **Invest in Ongoing Training and Support:**
   Implement continuous training programs to ensure all users, including patrons and administrators, are proficient in operating the system. Training sessions, user manuals, and access to online tutorials will enhance their confidence and efficiency in utilizing the system's features.

2. **Enhance Data Security Measures:**
   Strengthen data protection protocols by incorporating multi-factor authentication, encrypted data storage, and regular security audits to safeguard sensitive student and financial information from potential breaches.

3. **Improve System Performance and Stability:**
   Address any potential bottlenecks by optimizing database queries and upgrading infrastructure to handle peak periods of activity, such as at the beginning of academic years when membership updates are frequent.

4. **Expand Reporting and Analytics Capabilities:**
   Develop more advanced reporting tools to provide detailed insights into club membership trends, financial summaries, and activity success metrics. This will empower school management and patrons to make informed decisions.

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5. **Streamline Integration with School Systems:**
   Ensure seamless integration with other existing systems, such as the student information management system, to reduce redundancy, improve data accuracy, and enhance workflows.
6. **Regular Maintenance and Updates:**
   Schedule regular system maintenance and updates to address bugs, introduce new features, and adapt to emerging needs. Keeping the system updated will enhance reliability and longevity.
7. **Monitor and Evaluate System Performance:**
   Introduce monitoring tools to track performance metrics such as user satisfaction, system uptime, and error rates. This data can guide further system refinements and optimizations.
8. **Plan for Future Scalability:**
   Design a scalable architecture that can accommodate an increasing number of students, clubs, and activities as the school expands its operations or introduces new co-curricular offerings.

By implementing these recommendations, Victory School can maximize the effectiveness of its club membership system, enhance user satisfaction, and maintain a robust and efficient platform for managing student engagement in co-curricular activities.

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7.2 CONCLUSIONS

The review of the Victory School Club Membership System highlights the strengths, challenges, and opportunities associated with its implementation. This evaluation provides a roadmap for optimizing the system and ensuring it meets the evolving needs of the school community.

Key Takeaways:

- The automated system significantly improves the management of club memberships, finances, and events, reducing manual effort and enhancing decision-making.
- Initial challenges, such as user adaptation, data security risks, and system maintenance, need focused attention to ensure continued success.

- Recommendations include ongoing training, improved security, system optimization, and enhanced reporting to address identified weaknesses and maximize benefits.

Conclusion:
The Victory School Club Membership System represents a substantial improvement in managing the school's co-curricular activities, fostering student participation, and streamlining administrative tasks. By addressing the identified challenges and implementing the proposed recommendations, the school can ensure the system remains reliable, efficient, and adaptable to future requirements. Continuous monitoring and user feedback will be critical for maintaining system relevance and driving long-term success.

7.3 BIBLIOGRAPHY

The bibliography below shows a list of reference materials used in development.

1. Mburu S. & Chemwa G: Mastering Computer Studies; Project, Form 3 and 4 work.
2. Jeffrey L., et al: System Analysis and Design Methods, 5th-Ed, McGraw-Hill.
3. Mburu S. & Chemwa G: Longhorn Secondary Computer Studies Form 4, Longhorn Publishers, Nairobi.
4. Kendal E.K, Kendal J.E: System Analysis and Design, 5th Ed, Prentice-Hall, Kenya.
5. Connolly T and Beg C: Database Systems: A practical Approach to Design, Implementation and Management, 2nd-Ed Addison Wesley.

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APPENDIX

Ksh – Kenya Shillings

RAM – Random Access Memory

TB – Tera Byte

SQL – Structured Query Language

GB - Gigabyte

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